DEPARTMENT OF BIOELECTRONICS & BIOSENSORS

M. Sc. MATERIALS SCIENCE
REGULATIONS AND SYLLABUS

[For the candidates admitted from the Academic Year 2022 – 2023 onwards]

ALAGAPPA UNIVERSITY

(A State University Accredited with "A+" grade by NAAC (CGPA: 3.64) in the Third Cycle and Graded as Category-I University by MHRD-UGC)

, Karaikudi -630003, Tamil Nadu.

The panel of Members-Broad Based Board of Studies

Chairperson: Dr. C. Sekar, Professor And Head, Department of Bioelectronics and Biosensors, Alagappa University, Karaikudi. Areas of Expertise Materials Science: Metal Oxide Semiconductors, Carbon Nanostructures, Biomaterials, Low Dimensional Cuprates Sensors: Chemical Sensors, Biosensors for Medical, Food, Agricultural and Environmental Applications.	
Foreign Expert: Dr. Giovanni Neri, Professor, Department of Engineering, University of Messina. Italy. Areas of Expertise Catalysis, Gas Sensors, Biosensors.	
Indian Expert: Dr. J. Mathiyarasu, Principal Scientist, Biosensors Division, CSIR-Central Electrochemical Research Institute, India. Email: almathi@cecri.res.in. Areas of Expertise: Electrochemical Biosensors.	
Indian Expert: Dr. N. Rameshbabu, Associate Professor, Department of Metallurgy, National Institute of Trichy, Tamil nadu, India. Email: rameshrohith@gmail.com. Areas of expertise: Bio materials, Ceramic materials, Plasma electrolytic oxidation coatings	
Industry Expert: Dr. V. Viswabaskaran, The General Manager, VB ceramics, kottivakkam, Chennai, India. Email: drvbcrc@gmail.com, Area of Expertise: Ceramic Consultant	
Members: Dr. G. Ravi, Professor and Head, Department of Physics, Alagappa University, Karaikudi – 630003, Tamil Nadu, India. Email: raviganesa@rediffmail.com. Areas of Expertise: Crystal Growth of Organic & Inorganic Materials, Nano Materials Synthesis and Thin Films Preparation for Supercapacitors, Photocatalytic and Sensor Applications, Opto-Electronics and E-O Modulator–Devices	
Dr. V. Dharuman, Assistant Professor, Department of Bioelectronics and Biosensors, Alagappa University, Karaikudi – 630003, Tamil Nadu,India.Email:dharumanudhay@yahoo.com. Areas of Expertise: Chemistry/Electrochemistry/ Diabetic, Cancer Biosensors Development using, DNA, Antibody (Immunosensors) and Neurological Disorder Sensors.	
Dr. J. Wilson, Assistant Professor, Department of Bioelectronics & Biosensors, Alagappa University, Karaikudi – 630003, Tamil Nadu, India. Email:wilson.J2008@yahoo.com. Areas of Expertise: Conducting Polymers, Metal Oxides, Carbon Based Materials, Biosensors, and Lithium Batteries.	
Alumnus/Alumna: Mrs. S.Meenakshi, Research Scholar, Department of Bioelectronics and Biosensors, Karaikudi, India. Email: meenakshisolaimani@gmail.com. Areas of Expertise: Chemo-Biosensors, Shock waves.	

ALAGAPPA UNIVERSITY
DEPARTMENT OF BIOELECTRONICS & BIOSENSORS
Karaikudi -630003, Tamil Nadu.

REGULATIONS AND SYLLABUS-(CBCS-University Department)
[For the candidates admitted from the Academic Year 2022 – 2023 onwards]

Name of the Department: **Department of Bioelectronics & Biosensors**

Name of the Subject Discipline: Materials Science

Programme of Level: **M.Sc.,**

Duration for the Course: Full Time (Two Years)

1. **Choice-Based Credit System**

A choice-Based Credit System is a flexible system of learning. This system allows students to gain knowledge at their own tempo. Students shall decide on electives from a wide range of elective courses offered by the University Departments in consultation with the Department committee. Students undergo additional courses and acquire more than the required number of credits. They can also adopt an inter-disciplinary and intra-disciplinary approach to learning, and make the best use of the expertise of available faculty.

2. **Programme**

"Programme" means a course of study leading to the award of a degree in a discipline.

3. **Courses**

'Course' is a component (a paper) of a programme. Each course offered by the Department is identified by a unique course code. A course contains lectures/ tutorials/laboratory work/seminar/project work / practical training/report writing /Viva-voce, etc or a combination of these, to meet effectively the teaching and learning needs.

4. **Credits**

The Term "Credit" refers to the weightage given to a course, usually in relation to the instructional hours assigned to it. Normally in each of the courses credits will be assigned on the basis of the number of lectures/tutorial/laboratory and other forms of learning required to complete the course contents in a 15-week schedule. One credit is equal to one hour of lecture per week. For laboratory/field work one credit is equal to two hours.

5. **Semesters**

An Academic year is divided into two **Semesters.** In each semester, courses are offered in 15 teaching weeks and the remaining 5 weeks are to be utilized for conduct of examination and evaluation purposes. Each week has 30 working hours spread over 5 days a week.

6. **Medium of Instruction**

Medium of instruction is English

7. **Departmental committee**

The Departmental Committee consists of the faculty of the Department. The Departmental Committee shall be responsible for admission to all the programmes offered by the Department including the conduct of entrance tests, verification of records, admission, and evaluation. The Departmental Committee determine the deliberation of courses and specifies the allocation of credits semester-wise and course-wise. For each course, it will also identify the number of credits for lectures, tutorials, practicals, seminars etc. The courses (Core/Discipline Specific Elective/Non-Major Elective) are designed by teachers and approved by the Departmental Committees. Courses approved by the Departmental Committees shall be approved by the Board of Studies. A teacher offering a course will also be responsible for maintaining attendance and performance sheets (CIA - I, CIA-II, assignments and seminar) of all the students registered for the course. The Non-major elective programme and MOOCs coordinator are responsible for submitting the performance sheet to the Head of the department. The Head of the Department consolidates all such performance sheets of courses pertaining to the programmes offered by the department. Then forward the same to be Controller of Examinations.

8. **Programme General Objectives- (PGO) Minimum 6 objectives are required**

PGO-1	To educate basics of material science
PGO-2	To educate applications of materials in science
PGO-3	To motivate them to become researchers in materials advancements
PGO-4	To motivate them to understand interdisciplinary activities of materials
PGO-5	To enhance analytical skill by developing research problem and solving
PGO-6	To enhance critical thinking on societal issues in using materials

9. **Programme Specific Objectives-(PSO)- Minimum 6 objectives are required**

PSO-1	Learning applications of fundamental science in material design and development
PSO-2	Training in using analytical instruments application in material characterization
PSO-3	Educating on different forms of materials and their applications
PSO-4	Undertaking research problems in material science and solving
PSO-5	Effective design and implementation of materials in research and professional applications
PSO-6	

10. **Programme Outcome-(PO) - Minimum 6 objectives are required**

PO-1	Clear understanding of the core physical sciences related to material science
PO-2	Able to conduct experiments using materials based on requirements
PO-3	Identifying the research problems in material science
PO-4	Effective communication in terms of speaking, report writing and Presenting research problems
PO-5	Following professional and research ethics in handling materials
PO-6	Critical analysis and solving materials usage problems

11. Eligibility for admission

A candidate who has passed B.Sc., Degree Examination with Physics, or Chemistry or equivalent degree in Science with maths, physics or chemistry as allied subjects as University accepted by the syndicate as equivalent thereto.

12. Minimum Duration of programme

The programme is for a period of two years. Each year shall consist of two semesters viz. Odd and Even semesters. Odd semesters shall be from June / July to October / November and even semesters shall be from November / December to April / May. For each semester, there shall be 90 working days consisting of 6 teaching hours per working day (5 days/week).

13. Components

A PG programme consists of a number of courses. The term "course" is applied to indicate a logical part of the subject matter of the programme and is invariably equivalent to the subject matter of a "paper" in the conventional sense. The following are the various categories of the courses suggested for the PG programmes:

A. Core courses (CC)- "Core Papers" means "the core courses" related to the programme concerned including practicals and project work offered under the programme and shall cover Core competency, critical thinking, analytical reasoning, and research skill.
B. Discipline-specific electives (DSE) means the courses offered under the programme related to the major but are to be selected by the students, shall cover additional academic knowledge, critical thinking, and analytical reasoning.

C. Non-Major Electives (NME)- Exposure beyond the discipline

- Students have to undergo a total of two Non Major Elective courses with 2 credits offered by other departments (one in II Semester another in III Semester)
- A uniform time frame of 3 hours on a common day (Tuesday) shall be allocated for the Non-Major Electives
- Non Major Elective courses offered by the departments pertaining to a semester should be announced before the end of previous semester.
- Registration process: Students have to register for the Non-Major Elective course within 15 days from the commencement of the semester either in the department or NME portal (University Website).

D. Self Learning Courses from MOOCs platforms.

- MOOCs shall be on voluntary for the students.
- Students have to undergo a total of 2 Self Learning Courses (MOOCs) one in II semester and another in III semester.
- The actual credits earned through MOOCs shall be transferred to the credit plan of programmes as extra credits. Otherwise 2 credits/course be given if the Self Learning Course (MOOCs) is without credit.
- While selecting the MOOCs, preference shall be given to the course related to employability skills.
-

E. Projects / Dissertation /Internships (Maximum Marks: 200**)**

The student shall undertake the dissertation work during the fourth semester.

- **Plan of work**

<u>Project/Dissertation</u>

The candidate shall undergo Project/Dissertation Work during the final semester. The candidate should prepare a scheme of work for the dissertation/project and should get approval from the guide. The candidate, after completing the dissertation /project work, shall be allowed to submit it to the university departments at the end of the final semester. If the candidate is desirous of availing the facility from other departments/universities/laboratories/organizations they will be permitted only after getting approval from the guide and HOD. In such a case, the candidate shall acknowledge the same in their dissertation/project work.

- **Format to be followed for dissertation/project report**

The format /certificate for thesis to be followed by the student are given below

- Title page

- Certificate
- Acknowledgment
- Content as follows:

Chapter No	Title	Page number
1	Introduction	
2	Aim and objectives	
3	Review of literature	
4	Materials and methods	
5	Result	
6	Discussion	
7	Summary	
8	References	

- **Format of the title page**

Title of Dissertation/Project work

Dissertation submitted in partial fulfilment of the requirement for the degree of Master of Science in Material Science to the Alagappa University, Karaikudi -630003.

By

(Student Name)
(Register Number)
University Logo

Department of ----------

Alagappa University

(A State University Accredited with "A+" grade by NAAC (CGPA: 3.64) in the Third Cycle and Graded as Category-I University by MHRD-UGC, 2019: QS ASIA Rank-216, QS BRICS Rank-104,QS India Rank-20)

Karaikudi - 630003

(Year)

- **Format of certificates-**

Certificate -Guide

This is to certify that the thesis entitled "----------" submitted to Alagappa University, Karaikudi-630 003 in partial fulfilment for the degree of Master of Science in ---------- by Mr/Mis ----------(Reg. No:----------) under my supervision. This is based on the results of studies carried out by him/her in the Department of----------, Alagappa University, Karaikudi-630 003. This dissertation/Project or any part of this work has not been submitted elsewhere for any other degree, diploma, fellowship, or any other similar titles or record of any University or Institution.

Place: Karaikudi Research Supervisor

Date:_____

Certificate - (HOD)

This is to certify that the thesis entitled **"----------"** submitted by Mr/**Mis ----------(Reg No: ----------)** to the Alagappa University, in partial fulfilment for the award of the degree of **Master of** --------in ---------- is a bonafide record of research work done under the supervision of **Dr.----------,** Assistant Professor, Department of----------, Alagappa University. This is to further certify that the thesis or any part thereof has not formed the basis of the award to the student of any degree, diploma, fellowship, or any other similar title of any University or Institution.

Place: Karaikudi Head of the Department

Date:_____

Declaration (student)

I hereby declare that the dissertation entitled **"----------"** submitted to the Alagappa University for the award of the degree of Master of -------- in ---------- has been carried out by me under the guidance of **Dr.** ----------, Assistant Professor, Department of----------, Alagappa University, Karaikudi – 630 003. This is my original and independent work and has not previously formed the basis of the award of any degree, diploma, associateship, fellowship, or any other similar title of any University or Institution.

Place: Karaikudi **(----------)**

Date:_____

<u>Internship</u>

The students who have opted for an Internship must undergo industrial training in the reputed organizations to accrue industrial knowledge in the final semester. The student has to find industry related to their discipline (Public limited/Private Limited/owner/NGOs etc.,) in consultation with the faculty in charge and get approval from the head of the department and Departmental Committee before going for an internship.

- **Format to be followed for Internship report**

The format /certificate for internship report to be followed by the student are given below

- Title page -**Format of the title page**

Title of internship report

Internship report submitted in partial fulfilment of the requirement for the Master of degree in --------- to the Alagappa University, Karaikudi -630003.

By

(Student Name)

(Register Number)

University Logo

Department of ----------

Alagappa University

(A State University Accredited with "A+" grade by NAAC (CGPA: 3.64) in the Third Cycle and Graded as Category-I University by MHRD-UGC, 2019: QS ASIA Rank-216, QS BRICS Rank-104,QS India Rank-20)

Karaikudi - 630003

(Year)

- Certificate-**Format of certificate – faculty in-charge**

This is to certify that the report entitled "----------" submitted to Alagappa University, Karaikudi-630 003 in partial fulfilment for the Master of Science in ----------by Mr/Mis---------- (Reg. No:----------) under my supervision. This is based on the work carried out by him/her in the organization M/S ----------. This Internship report or any part of this work has not been submitted elsewhere for any other degree, diploma, fellowship, or any other similar record of any University or Institution.

Place: Research Supervisor

Date:_____

<u>Certificate (HOD)</u>

This is to certify that the Internship report entitled **"----------"** submitted by Mr/Mis**.----------(Reg No:----------)** to the Alagappa University, in partial fulfilment for the award of the Master of Science in **-------** is a bonafide record of Internship report done under the supervision of ----------, Assistant Professor, Department of ----------, Alagappa University and the work carried out by him/her in the organization M/S ----------. This is to further certify that the thesis or any part thereof has not formed the basis of the award to the student of any degree, diploma, fellowship, or any other similar title of any University or Institution.

Place: Karaikudi Head of the Department

Date:_____

- Certificate-**Format of certificate – Company supervisor or Head of the Organization**

This is to certify that the report entitled "----------" submitted to Alagappa University, Karaikudi-630 003 in partial fulfilment for the Master of Science in ----------by Mr/Mis---------- (Reg. No:----------) under my supervision. This is based on the work carried out by him/her in our organization M/S ---------- for the period of three months or ------. This Internship report or any part of this work has not been submitted elsewhere for any other degree, diploma, fellowship, or any other similar record of any University or Institution.

Place: Supervisor or in charge

Date:_____

Declaration (student)

I hereby declare that the dissertation/project entitled **"----------"** submitted to the Alagappa University for the award of the **Master of Science in** ---------- has been carried out by me under the supervision of----------, Assistant Professor, Department of----------, Alagappa University, Karaikudi – 630 003. This is my original and independent work carried out by me in the organization M/S ---------- for the period of three months or ------ and has not previously formed the basis of the award of any degree, diploma, associateship, fellowship, or any other similar title of any University or Institution.

Place: Karaikudi **(----------)**

Date:_____

- Acknowledgment

- Content as follows:

Chapter No	Title	Page number
1	Introduction	
2	Aim and objectives	
3	Organisation profile /details	
4	Method work	
5	Observation and knowledge gained	
6	Summary and outcome of the study	
7	References	

- **No. of copies of the dissertation/project report/internship report**

The candidate should prepare three copies of the dissertation/project/report and submit the same for the evaluation of examiners. After evaluation, one copy will be retained in the department library, one copy will be retained by the guide and the student shall hold one copy.

14. Teaching methods

Department teachers shall instruct using power point presentation along with white board and smart boards. Use of internet and library as and when required for effective learning

15. Attendance

Students must have earned 75% of attendance in each course for appearing for the examination. Students who have earned 74% to 70% of attendance need to apply for condonation in the prescribed form with the prescribed fee. Students who have earned 69% to 60% of attendance need to apply for condonation in the prescribed form with the prescribed fee along with the Medical Certificate. Students who have below 60% of attendance are not eligible to appear for the End Semester Examination (ESE). They shall re-do the semester(s) after completion of the programme

16. Examination

The examinations shall be conducted separately for theory and practical's to assess (remembering, understanding, applying, analysing, evaluating, and creating) the knowledge required during the study. There shall be two systems of examinations viz., internal and external examinations. The internal examinations shall be conducted as Continuous Internal Assessment tests I and II (CIA Test I & II).

A. Internal Assessment

The internal assessment shall comprise a maximum of 25 marks for each subject. The following procedure shall be followed for awarding internal marks.

Theory -25 marks

Sr.No	Content	Marks
1	Average marks of two CIA test	15
2	Attendance	2
3	Seminar/group discussion/quiz	4
4	Assignment/field trip report/case study report	4
	Total	25

Practical -25 Marks

1	Major Experiment	10 marks
2	Minor Experiment	5 marks
3	Spotter (2x 5/ 4 x4) or any other mode	10 marks
	Total	25 Marks

Project/Dissertation/internship-50 Marks (assess by Guide/incharge/supervisor)

1	Two presentations (mid-term)	30 Marks
2	Progress report	20 Marks
	Total	50 Marks

B. External Examination

- There shall be examinations at the end of each semester, for odd semesters in the month of October / November; for even semesters in April / May.
- A candidate who does not pass the examination in any course(s) may be permitted to appear in such failed course(s) in the subsequent examinations to be held in October / November or April / May. However candidates who have arrears in Practical shall be permitted to take their arrear Practical examination only along with Regular Practical examination in the respective semester.
- A candidate should get registered for the first semester examination. If registration is not possible owing to shortage of attendance beyond condonation limit / regulation prescribed OR belated joining OR on medical grounds, the candidates are permitted to move to the next semester. Such candidates shall re-do the missed semester after completion of the programme.
- For the Project Report/ Dissertation Work / internship the maximum marks will be 100 marks for project report evaluation and for the Viva-Voce it is 50 marks (if in some programmes, if the project is equivalent to more than one course, the project marks would be in proportion to the number of equivalent courses).
- Viva-Voce: Each candidate shall be required to appear for Viva-Voce Examination (in defense of the Dissertation Work /Project/ internship).

C. Scheme of External Examination (Question Paper Pattern)

Theory - Maximum 75 Marks

Section A	10 questions. All questions carry equal marks. (Objective type questions)	10 x 1 = 10 Marks	10 questions – 2 each from every unit
Section B	5 questions Either / or type like 1.a (or) b. All questions carry equal marks and each answer should not exceed one page or 250 words.	5 x 5 = 25	5 questions – 1 each from every unit
Section C	Essay type questions 5 out of 8 questions. All questions carry equal marks and each answer should not exceed two page	5 x8 = 40	5 question –Should cover all units

Practical –Maximum 75 Marks

Section A	Major experiment	15 Marks
Section B	Minor experiment	10 Marks
Section C	Experimental setup	5 Marks
Section D	Spotters (5 x5)	25 Marks
Section E	Record note	10 Marks
Section F	Vivo voce	10 Marks

17. Dissertation /Project report/Internship report Scheme of evaluation

Dissertation /Project report/Internship report	100 Marks
Vivo voce	50 Marks

18. Results

The results of all the examinations will be published through the Department where the student underwent the course as well as through University Website

19. Passing minimum

- A candidate shall be declared to have passed in each course if he/she secures not less than 40% marks in the End Semester Examinations and 40% marks in the Internal Assessment and not less than 50% in the aggregate, taking Continuous assessment and End Semester Examinations marks together.
- The candidates not obtained 50% in the Internal Assessment are permitted to improve their Internal Assessment marks in the subsequent semesters (2 chances will be given) by writing the CIA tests and by submitting assignments.
- Candidates, who have secured the pass marks in the End-Semester Examination and in the CIA but failed to secure the aggregate minimum pass mark (E.S.E + C I.A), are permitted to improve their Internal Assessment mark in the following semester and/or in University examinations.
- A candidate shall be declared to have passed in the Project Work if he /she gets not less than 40% in each of the Project Report and Viva-Voce and not less than 50% in the aggregate of both the marks for Project Report and Viva-Voce.
- A candidate who gets less than 50% in the Project Report must resubmit the Project Report. Such candidates need to take again the Viva-Voce on the resubmitted Project.

20. Grading of the Courses

The following table gives the marks, Grade points, Letter Grades and classifications meant to indicate the overall academic performance of the candidate.

Conversion of Marks to Grade Points and Letter Grade (Performance in Paper / Course)

RANGE OF MARKS	GRADE POINTS	LETTER GRADE	DESCRIPTION
90 - 100	**9.0 – 10.0**	**O**	**Outstanding**
80 - 89	**8.0 – 8.9**	**D+**	**Excellent**
75 - 79	**7.5 – 7.9**	**D**	**Distinction**
70 - 74	**7.0 – 7.4**	**A+**	**Very Good**
60 - 69	**6.0 – 6.9**	**A**	**Good**
50 - 59	**5.0 – 5.9**	**B**	**Average**
00 - 49	**0.0**	**U**	**Re-appear**
ABSENT	**0.0**	**AAA**	**ABSENT**

a) Successful candidates passing the examinations and earning GPA between 9.0 and 10.0 and marks from 90 – 100 shall be declared to have Outstanding (O).
b) Successful candidates passing the examinations and earning GPA between 8.0 and 8.9 and marks from 80 - 89 shall be declared to have Excellent (D+).
c) Successful candidates passing the examinations and earning GPA between 7.5 – 7.9 and marks from 75 - 79 shall be declared to have Distinction (D).
d) Successful candidates passing the examinations and earning GPA between 7.0 – 7.4 and marks from 70 - 74 shall be declared to have Very Good (A+).
e) Successful candidates passing the examinations and earning GPA between 6.0 – 6.9 and marks from 60 - 69 shall be declared to have Good (A).
f) Successful candidates passing the examinations and earning GPA between 5.0 – 5.9 and marks from 50 - 59 shall be declared to have Average (B).
g) Candidates earning GPA between 0.0 and marks from 00 - 49 shall be declared to have Re-appear (U).
h) Absence from an examination shall not be taken as an attempt.

From the second semester onwards the total performance within a semester and continuous performance starting from the first semester are indicated respectively **by Grade Point Average (GPA) and Cumulative Grade Point Average (CGPA).** These two are calculated by the following formulate

GRADE POINT AVERAGE (GPA) = $\Sigma_i C_i \; G_i / \Sigma_i C_i$

$$\text{GPA} = \frac{\text{Sum of the multiplication of Grade Points by the credits of the courses}}{\text{Sum of the credits of the courses in a Semester}}$$

21. Classification of the final result

CGPA	Grade	Classification of Final Result
9.5 – 10.0 9.0 and above but below 9.5	**O+** **O**	First Class – Exemplary*
8.5 and above but below 9.0 8.0 and above but below 8.5 7.5 and above but below 8.0	**D++** **D+** **D**	First Class with Distinction*
7.0 and above but below 7.5 6.5 and above but below 7.0 6.0 and above but below 6.5	**A++** **A+** **A**	First Class
5.5 and above but below 6.0 5.0 and above but below 5.5	**B+** **B**	Second Class
0.0 and above but below 5.0	**U**	Re-appear

The final result of the candidate shall be based only on the CGPA earned by the candidate.

a) Successful candidates passing the examinations and earning CGPA between 9.5 and 10.0 shall be given Letter Grade (O+), those who earned CGPA between 9.0 and 9.4 shall be given Letter Grade (O) and declared to have First Class –Exemplary*.
b) Successful candidates passing the examinations and earning CGPA between 7.5 and 7.9 shall be given Letter Grade (D), those who earned CGPA between 8.0 and 8.4 shall be given Letter Grade (D+), those who earned CGPA between 8.5 and 8.9 shall be given Letter Grade (D++) and declared to have First Class with Distinction*.
c) Successful candidates passing the examinations and earning CGPA between 6.0 and 6.4 shall be given Letter Grade (A), those who earned CGPA between 6.5 and 6.9 shall be given Letter Grade (A+), those who earned CGPA between 7.0 and 7.4 shall be given Letter Grade (A++) and declared to have First Class.

d) Successful candidates passing the examinations and earning CGPA between 5.0 and 5.4 shall be given Letter Grade (B), those who earned CGPA between 5.5 and 5.9 shall be given Letter Grade (B+) and declared to have passed in Second Class.

e) Candidates those who earned CGPA between 0.0 and 4.9 shall be given Letter Grade (U) and declared to have Re-appear.

f) Absence from an examination shall not be taken as an attempt.

CUMULATIVE GRADE POINT AVERAGE (CGPA) = $\Sigma_n \Sigma_i C_{ni} \quad G_{ni} / \Sigma_n \Sigma_i C_{ni}$

$$\text{CGPA} = \frac{\text{Sum of the multiplication of Grade Points by the credits of the entire Programme}}{\text{Sum of the credits of the courses for the entire Programme}}$$

Where '**Ci**' is the Credit earned for Course i in any semester; '**Gi**' is the Grade Point obtained by the student for Course i and 'n' refers to the semester in which such courses were credited.

CGPA (Cumulative Grade Point Average) = Average Grade Point of all the Courses passed starting from the first semester to the current semester.

Note: * The candidates who have passed in the first appearance and within the prescribed Semesters of the PG Programme are alone eligible for this classification.

20. Maximum duration of the completion of the programme

The maximum period for completion of **M.Sc.,/ M.A.,/ M.B.A/ B.Ed.,/ M.Ed.,/ B.P.Ed/ M.P.Ed** in ---------- shall not exceed eight semesters continuing from the first semester.

21. Conferment of the Master's Degree

A candidate shall be eligible for the conferment of the Degree only after he/ she has earned the minimum required credits for the Programme prescribed there for (i.e. 90 credits). Programme).

22. Village Extension Programme

The Sivaganga and Ramnad districts are very backward districts where a majority of people Lives in poverty. The rural mass is economically and educationally backward. Thus the aim of the introduction of this Village Extension Programme is to extend out to reach environmental awareness, social activities, hygiene, and health to the rural people of this region. The students in their third semester have to visit any one of the adopted villages within the jurisdiction of Alagappa University and can arrange various programs to educate the rural mass in the following areas for three day based on the theme.1. Environmental awareness 2. Hygiene and Health. A minimum of two faculty members can accompany the students and guide them.

DEPARTMENT OF BIOELECTRONICS & BIOSENSORS

PG –Materials Science

S. No	Paper Code	Title of the paper		T/P	Credits	Hours/ Week	Marks		
		I Semester					**I**	**E**	**Total**
1	542101	Core 1	Materials Physics	T	5	5	25	75	100
2	542102	Core 2	Thermodynamics	T	5	5	25	75	100
3	542103	Core 3	Electronics and Instrumentation	T	4	4	25	75	100
4	542104	Core 4	Electromagnetic Theory and Optics	T	4	4	25	75	100
6	542105	Core 5	Lab-I : Electrical and Optical Characterization	P	4	8	25	75	100
7	542501	Discipline Specific Elective (DSE) – I		T	3	3	25	75	100
		Library / Yoga/ Counselling/Field trip				1			
					25	**30**	**150**	**450**	**600**
		II Semester							
8	542201	Core 6	Materials Chemistry	T	4	4	25	75	100
9	542202	Core 7	Characterization of Materials	T	4	4	25	75	100
10	542203	Core 8	Quantum Mechanics	T	4	4	25	75	100
11	542204	Core 9	Crystal Growth	T	4	4	25	75	100
12	542205	Core 10	Lab-II: Materials synthesis and Characterization	P	4	8	25	75	100
13		Discipline Specific Elective(DSE) - II		T	3	3	25	75	100
14	542701	Non-Major Elective :1		T	2	3	25	75	100
15		Self-learning course (SLC) –MOOCs**			Extra credit				
					25	**30**	**175**	**525**	**700**
		III Semester							
15	542301	Core 11	Nanomaterials	T	4	4	25	75	100
16	542302	Core 12	Polymer and Composite Materials	T	4	4	25	75	100
17	542303	Core 13	Solid State Physics	T	4	4	25	75	100
18	542304	Core 14	Ceramic materials	T	4	4	25	75	100
19	542305	Core 15	Lab-III: Device Characterization	P	4	8	25	75	100
20		Discipline Specific Elective(DSE) – III		T	3	3	25	75	100
21		Non-Major Elective:2		T	2	3	25	75	100
22		Self-learning course (SLC) –MOOCs**			Extra credit				
					25	**30**	**175**	**525**	**700**
		IV Semester							
23	542999	Core 16	***Project Dissertation Work or Internship programme	P	15	30	50	150	200
					15	**30**	**50**	**150**	**200**
			Total		**90 +**		**550**	**1650**	**2200**

*DSE – Discipline Specific Elective.

**SLC- Voluntary basis

*** Dissertation / internship report –Marks -Vivo-voce (50) + thesis (100) + internal (50) = 200

T-Theory

P-Practical

Non-Major Elective-Courses offered to the other Department

Sl. No.	Paper Code	Semester	Title of the paper	Crédits	Hours/ Week	Marks		
						I	E	T
1	542701	II	Electronics for Daily Life	2	3	25	75	100
		II	Food Chemistry	2	3	25	75	100
2	542702	III	Nanomaterial Bio Sensors	2	3	25	75	100
		III	Green Chemistry	2	3	25	75	100

DISCIPLINE SPECIFIC ELECTIVE (DSE)

No.	Code	Title of the Course	No. of Credits	Contact Hours
I SEMESTER				
1	542501	Biomaterials	3	3
2	542502	Molecular Electronics	3	3
3	542503	Non-Destructive Testing	3	3
4	542504	Nonlinear Optics and Materials	3	3
5	542505	Laser and Applications	3	3
6	542506	Python Programming	3	3
II SEMESTER				
7	542507	Molecular spectroscopy	3	3
8	542508	Semiconductor Materials And Devices	3	3
9	542509	Physical Metallurgy	3	3
10	542510	Materials Processing	3	3
11	542512	Corrosion Science And Engineering	3	3
12	542513	Solid State Ionics	3	3
III SEMESTER				
13	542514	Biosensors	3	3
14	542515	Bioelectronics	3	3
15	542516	Chemical Sensors	3	3
16	542517	Thin Film Science And Technology		
17	542519	Superconducting Materials and Applications	3	3

M.Sc. Materials Science

(Regular Programme)

Curriculum (2022-2023 onwards)

under

Choice Based Credit System

I Semester							
CC		**Core -1**	**L**	**T**	**P**	**C**	**H/W**
Course code	**542101**	**MATERIALS PHYSICS**	**-**	**T**	**-**	**5**	**5**
Objectives	To impart knowledge on various properties of materials • To introduce the concepts of various mechanical test and plastic deformation • To introduce various dielectric materials and their application • To expose different types of magnetic materials their properties and applications. • To study the optical properties, LED and LCD and their • To become familiar with smart materials, shape memory alloys, nanomaterials						
Unit I	**MECHANICAL PROPERTIES -** Plastic deformation by slip – the shear strength of perfect and real crystals -dislocation movement– methods of strengthening against plastic yield – Creep – mechanisms – fracture – ductile fracture – brittle fracture – Griffith criterion – fracture toughness – fatigue fracture - mechanical tests - tensile, hardness and creep tests.						
Unit II	**DIELECTRIC PROPERTIES -** Dielectric constant and polarizability - different kinds of polarization - Internal electric field in a dielectric – Clausius - Mossotti equation - dielectric in a ac field - dielectric loss - ferroelectric - types and models of ferro electric transition - electrets and their applications – piezoelectric and pyroelectric materials.						
Unit III	**MAGNETIC PROPERTIES -**. Classification of magnetic materials- origin of magnetism – Langevin and Weiss theories - exchange interaction - magnetic anisotropy - magnetic domains - molecular theory – hysterisis - hard and soft magnetic materials - ferrite structure and uses.						
Unit IV	**OPTICAL PROPERTIES -** Optical absorption in insulators, semiconductors and metals – band to band absorption – luminescence - photoconductivity. Injection luminescence and LEDs- LED materials - super luminescent LED materials - liquid crystals - properties and structure - liquid crystal displays-comparison between LED and LC displays.						
Unit V	**ADVANCED MATERIALS -** Metallic glasses - preparation, properties and applications - SMART materials - piezoelectric, magnetostrictive, shape memory alloys - rheological fluids - CCD device materials and applications - solar cell materials						
Outcome	After completing the course, the students shall have • knowledge in mechanical tests and plastic deformation mechanism • know the application and various properties of dielectric materials • understood the fundamentals on magnetic materials properties and their application • knowledge on optical properties and their applications • understood about advanced materials and preparation methods						

Reference and Text Books: -

1. Kasap S.O. (2019). *Principles of Electronic Materials and Devices*. McGraw-Hill Education.
2. Otsuka K. and Wayman C.M. (1998). *Shape Memory Materials*, Cambridge University Press.
3. Raghavan V. (2015). *Materials Science and Engineering: A First Course*. PHI Learning.
4. Suryanarayana C.and Inoue A. (2017). *Bulk Metallic Glasses*, CRC Press.

I Semester							
CC		**Core 2**	**L**	**T**	**P**	**C**	**H/W**
Course code	**542102**	**THERMODYNAMICS**	**-**	**T**	**-**	**5**	**5**

Objectives	To make the students understand • Mechanical aspects of systems and mathematical methods of Classical Mechanics. • The concepts laws of thermodynamics its applications and phase equilibria. • The statistical mechanics of systems, probability distribution laws. • The applications of statistical thermo dynamical methods in solid state physics. • The basic concepts of heat and mass transfer and its application in hydrodynamics.
Unit I	**LAGRANGIAN AND HAMILTONIAN DYNAMICS-**Mechanics of single and system of particles - Conservation laws – Constraints - Generalised coordinates -Virtual work - D'Alembert's principle – Lagranges equation of motion– Cyclic co-ordinates - Hamilton's equations of motion -Euler Lagrange equation - Principle of least action.
Unit II	**CANONICAL TRANSFORMATION, BRACKETS AND RIGID BODY-** Canonical transformation – Generating functions - Poisson brackets - Lagrange brackets - Relation between Lagrange and Poisson brackets-Jacobi Idendity. Rigid body dynamics: Euler's angles- -Angular velocity – Principal moment of inertia -Kinetic energy.
Unit III	**THERMODYNAMICS-** Laws of thermodynamics- internal energy- Enthalpy- Entropy- Helmholtz and Gibbs free energies – Thermodynamic relations – Euler equation – Maxwell's relations and applications – Chemical Potential- Gibbs phase rule – phase equilibria (single and multicomponent systems) - Clausius – Clayperon equation – law of mass action – first order phase transition in single component systems – Second order phase transition.
Unit IV	**CLASSICAL AND QUANTUM STATISTICS -** Micro and Macro States - Ensembles - Microcanonical, canonical and grand canonical ensembles – Maxwell – Boltzmann, Bose-Einstein and Fermi-Dirac statistics – Comparison of MB, BE and FD statistics.
Unit V	**APPLICATION OF STATISTICS -** Planck's Radiation law – Stefan-Boltzmann law – Einstein model of a solid – Bose condensation – Classical partition function and classical ideal gas – Equipartition theorem – Semiconductor statistics – Statistical equilibrium of electrons in semiconductors.
Outcome	After the completion of the course, students • become familiar with theories of single and multiple particle systems • understand the transformation of particles. • knowledge on thermodynamics and its applications • get familiar with classical and quantum statistics and its types • shall gain knowledge on application of statistics in practice.

Reference and Text Books: -

1. Callen.H.B. (1966).*Thermodynamics*. John Wiley and Sons.
2. Engel. T. andReid.P. (2007). *Thermodynamics, Statistical Thermodynamics & Kinetics*, Pearson Edu.
3. Goldstein. H. PooleC.P. andSafko.J. (2017). *Classical Mechanics*. Pearson Education, Inc.
4. Gupta.M.C. (1993). *Statistical Thermodynamics*. Wiley Eastern Ltd.
5. Holman.J.P. (2008). *Heat transfer*. Tata McGraw Hill.
6. RanaN.C. andJoag.P.S. (2017). *Classical Mechanics*. McGraw Hill Education.
7. Reif.F. (2010). *Fundamentals of Statistical and Thermal Physics*. Waveland Press.
8. UpadhyayaJ. C.. (2005).*Classical Mechanics*. Himalaya Publishing House.

I Semester							
CC		**Core-3**	**L**	**T**	**P**	**C**	**H/W**
Course code	**542103**	**ELECTRONICS AND INSTRUMENTATION**	**-**	**T**	**-**	**4**	**4**
Objectives	Objectives of the programme is to make the students • understand the concept of analog electronics. • gain knowledge in advanced concepts of digital electronics. • learn concepts of optoelectronics. • understand knowledge in designing electronic instruments. • study basic concepts of nanoelectronics in materials perspective .						
Unit I	**ANALOG ELECTRONICS -**Operational amplifiers: Introduction –op-amp parameters– feedback– differential amplifier –comparators – mathematical operations – active filters – instrumentation amplifiers – isolation amplifiers – OTAs –Voltage regulators: Principles and operations.						
Unit II	**DIGITAL ELECTRONICS-** Introduction – overview of logic functions and logic gates – combinational logic – flip-flops and related circuits – sequential logic – registers, counters, shift-registers and memory – microprocessor architecture – A/D and D/A conversion.						
Unit III	**OPTOELECTRONICS-** LEDs – semiconductor lasers – photodiodes – solar cells – photodetectors – optical fibers – communication – optoelectronic modulation and switching devices – optocoupler– optical data storage devices.						
Unit IV	**ELECTRONIC INSTRUMENTATION -** Basics of instrumentation system – transducers – types of transducers – strain gauges – RTDs – LVDT – piezoelectric transducers – load cell – flow meters – signal conditioning – data acquisition and conversion – data transmission.						
Unit V	**NANOELECTRONICS -** MOSFETs - `electron transport in nanostructures - resonant tunneling diodes – single electron transfer devices – molecular switches and memory storage – nano-electromechanical systems - quantum dot cellular automata.						
Outcome	After the completion of the course, students become knowledgeable in • analog amplifiers and their functions • digital amplifiers and their functions • optical applications of semiconductors and their functions • material applications in instrumental development • recent advancement in transistor applications						

Reference and Text Books: -

Bhattacharya P. (2019). *Semiconductor Optoelectronic Devices*. Pearson Education.
Chua L.O. Desoer C. AandKuh E.S. (1997).*Linear and Nonlinear Circuits*. McGraw-Hill.
Cooper W.D.(1991).*Electronic Instrumentation and Measurement Techniques*. Prentice Hall of India.
Floyd.T.L. (2015).*Electronic devices*. Pearson Education.
Hanson G.W. (2009). *Fundamentals of Nanoelectronics*. Pearson Education Inc.
Horowitz Pand Hill W. (2006). *Art of electronics*. Cambridge Univ. Press.
Kalsi H.S. (2017). *Electronic Instrumentation*. McGraw Hill Education.
Lakshmanan Mand Murali K. (1996). *Chaos in Nonlinear Oscillators*. World Scientific.
Malvino A.P. (2011). *Electronic principles*. Tata McGraw-Hill.

I Semester							
CC		**Core-4**	**L**	**T**	**P**	**C**	**H/W**
Course code	**542104**	**ELECTROMAGNETIC THEORY AND OPTICS**	**-**	**T**	**-**	**4**	**4**

Objectives	The objectives of the course is to introduce • Fundamentals of Maxwell's equations are their applications in different situations. • Insight on fundamental laws of optics and its relation with Maxwell's Equations. • novel calculus of tensors and illustrate their usage in different Material properties. • various optical activities and their applications in material characterization. • basics of non-linear optical effects and non-linear optical materials.
Unit I	**MAXWELL'S EQUATIONS -**Review of Gauss's law in electrostatics and magnetism - Ampere's law- Faraday's law -displacement current - Maxwell's equations - differential and integral forms - scalar and vector potentials and applications Potential due to a uniformly charged sphere - magnetic induction due to a current carrying wire.
Unit II	**ELECTROMAGNETIC WAVE PROPAGATION -** Plane electromagnetic waves in free surface - Poynting vector - characteristic impedance - wave equation in an isotropic medium - wave equation in insulators and conductors - reflection by a perfect conductor - normal and oblique incidence - Fresnel equations for parallel and perpendicular polarization.
Unit III	**CRYSTAL OPTICS -** Crystal symmetry-Light propagation in anisotropic media – Maxwell's equations: the constitutiverelation -Index ellipsoid – wave plates – Biaxial media: Optic axes – positive and negative crystals - Electrical conductivity tensor- - stress optic tensors - third rank tensors – Linear Electro-optic effect - Fourth rank tensors: third order susceptibility tensor.
Unit IV	**OPTICAL ACTIVITY-** Optical Polarization – Magneto-optical effects – Magneto-optical Kerr and Faraday effect - Kerr and Pockel effect - applications - Harmonics and sum & frequency generation - stimulated Brillouin scattering (SBS) - stimulated Raman scattering (SRS) – applications of SBS and SRS for material characterization – examples.
Unit V	**NONLINEAR OPTICS -** Theory and applications of non-linear effects - frequency conversion - optical switching - phase conjugation - optical bistability - nonlinear optical materials - NLO crystals, properties and applications.
Outcome	After completion of this paper the students shall understand the effect of light propagation in materials and how materials change the nature of electromagnetic wave. Specifically they will be able to: • Derive Maxwell's equations and apply them to study the electrostatics and magneto statics. • Understand boundary conditions between different materials and reflection and refraction of light based on Maxwell's equations. • Appreciate the use of tensors in determining crystal symmetry and in explaining advanced properties of materials like elastic properties, piezoelectric effect, eletro-optic effect etc. • Elucidate how optical activities occur in materials and how they can be used to further characterize materials. • Apprehend the fundamentals of Non-linear optical effects, the nature of materials exhibiting such properties, and their applications.

Reference and Text Books: -

1. Corson D. and Lorrain P. (2013).*Introduction to Electromagnetic Fields and Waves*, Literary Licensing, LLC.
2. Fleisch D. (2008). *A student's Guide to Maxwell's Equations.* Cambridge University Press.
3. Griffiths D.J. (2015). *Introduction to Electrodynamics*. Pearson Education.
4. Jordan E.Cand Balmain K.G (2015). *Electromagnetic Waves and Radiating Systems*. Pearson Education.
5. New G. (2014). *Introduction to Nonlinear Optics*, Cambridge University Press.
6. Nye J.F. (1997) *Physical Properties of Crystals*. Oxford University Press.
7. Yariv A. and Yeh P. (2007). *Photonics.* Oxford University Press.

I Semester							
CC		**Core -5**	**L**	**T**	**P**	**C**	**H/W**
Course code	**542105**	**LAB I Electrical and Optical Characterization**	**-**	**-**	**P**	**4**	**8**
Objectives	The main objective of the course is to impart knowledge on design and methodology for investigation of various optical and electrical properties of materials. In addition, the course is designed • To make the students realise the importance of investigating material propertis • To introduce the concepts of various optical techniques. • To introduce the students about various electrical materials and their application. • To expose the students to magnetic materials and their properties. • To educate electrochemical behaviour of materials and their characterization						
Practicals	1. Optical band gap determination. UV-visible spectrometric method 2. Photoluminescence properties of metal oxides 3. UV – visible characterization of polymers 4. Fluorescence spectral behavior of organic polymers 5. Fluorescence properties of carbon materials 6. Fluorescence properties of metal composites 7. Electrical conductivity of metals and alloys with temperature-four probe method. 8. Hall effect - Determination of Hall co-efficient, charge carrier density and mobility. 9. Magnetic susceptibility-Quincke's method. 10. Crystal Growth-Solution technique. 11. Crystal Growth-Gel technique. 12. Ferroelectricity – Hysteresis loop - coercivity, retentivity and saturation magnetisation. 13. Fraunhofer diffraction - using laser 14. Cyclic voltammetric characterization of electrode materials 15. Electrochemical impedance characterization of electrode/electrolyte interface 16. Chronoamperometric technique for analyte detection using sensor materials 17. Differntial pulse voltammetric analysis of analyte using sensor materials						
Outcome	After the completion of the course, students shall be familiar with various experiments to investigate the optical and electrical properties of materials. With the hand on training, they will the importance of investigating material properties to decide about the materials quality and potential application o of the materials.						

II SEMSESTER

II Semester							
CC		**Core 6**	**L**	**T**	**P**	**C**	**H/W**
Course code	**542201**	**MATERIALS CHEMISTRY**	**-**	**T**	**-**	**4**	**4**
Objectives	The overarching goals of introducing materials chemistry course is • to elucidate fundamental chemical aspects of materials' composition-structure-property relationships • to know various types of materials based on bondings • to understand the structure and properties of polymers • to know the material preparation techniques • to provide the knowledge needed to design and produce materials with tailored properties from first principles.						
Unit I	**BONDING IN SOLIDS** Introduction Bonding in Elemental Solids, Covalent Bonding, Metallic Bonding, van der Waals Bonding, Bonding in Multielement Crystals. Ionic Bonding, Mixed Ionic–Covalent Bonding and Ionicity, Hydrogen Bonding, Cohesive Energies, Summary of Some Atomic Properties and Parameters Ionization Energy and Electron Affinity, Electronegativity, Atomic Radii: Ionic, Covalent, Metallic, and van der Waals. Problems						
Unit II	**CLASSES OF MATERIALS** Characteristic Properties of Semiconductors- Microscopic Properties. Energy-Band Structure and Energy Gaps. Dynamics of Electron Motion. Excited States of Electrons. Doping and Defects Dimensionality and Quantum Confinement Macroscopic Properties. Electrical Conductivity and Mobility. Effects of Magnetic Fields. Optical Properties. Examples of Semiconductors - Elemental Semiconductors and Their Compounds and Alloys. Compound Semiconductors and Their Alloys; Applications of Semiconductors.						
Unit III	**POLYMERS:** Structure and Geometry of Polymers. Polymer Crystals. Defects in Polymers Mechanical Properties. Polymers Under Tension. Viscoelasticity Thermal Properties. Thermal Properties of Polymers Applications. Structural Plastics. Polymeric Ionic Conductors. Photoresists Piezoelectric Polymers. Liquid Crystals						
Unit IV	**SURFACE AND THIN FILMS INTERFACES:** Ideal Surfaces. Real Surfaces. Relaxation. Reconstruction. Surface Defects Electronic Properties of Surfaces. Work Function Thermionic Emission Field Emission Photoemission Surface States Surface Modification Anodization Passivation Surface Phonons Surface Processes Adhesion and Friction Surface Plasmons Dispersion Forces. Friction						
Unit V	**SYNTHESIS AND PROCESSING OF MATERIALS:** Introduction Issues in Synthesis and Processing. Thermodynamic and Chemical Effects. Kinetic Effects. Crystal Growth. Synthesis and Processing of Metals. Synthesis and Processing of Steels. Synthesis and Processing of Stainless Steels Synthesis and Processing of Ceramics and Glasses. Powder Synthesis. Sol–Gel Synthesis Synthesis and Processing of Polymers and Carbon Molecules. Polymerization. Catalysts in Polymer Synthesis. Synthesis of Carbon Nanotubes						
Outcome	After completion of the course, the students shall be able to • recognize different bonding nature forming crystals or materials in their structure. • familiarize with classes of material based on their physical and electronic properties. • gain knowledge on polymer properties pertaining to material applications. • understand the interaction of thin films with material surfaces. • synthesise materials for various applications.						

Reference and Text Books: -

1. Joel I. Gersten Frederick W. Smith The Physics And Chemistry Of Materials,
2. John Wiley & Sons, Inc., ISBN 0-471-05794-0 , 2001
3. Leonard V. Interrante, editor, Lawrence A. Casper, editor, Arthur B. Ellis, editor, Materials chemistry: an emerging discipline, Advances in chemistry series, ISSN 0065-2393; 245) ISBN 0-8412-2809-4
4. Bradley D. Fahlman, Materials Chemistry, Published by Springer, ISBN 978-1-4020-6120-2 (e-book)
5. Robert J. Naumann, Introduction to the physics and chemistry of materials, CRC Press13: 978-1-4200-6134-5, 2008

II Semester							
CC		**Core-7**	**L**	**T**	**P**	**C**	**H/W**
Course code	**542202**	**CHARACTERISATON OF MATERIALS**	**-**	**T**	**-**	**4**	**4**
Objectives	The objective of the course is to make the students • understand some important thermal analysis techniques. • familiarize with image formation in an optical microscope and learn other specialized microscopic techniques. • learn the principle of working of electron microscopes and scanning probe microscopes. • understand some important semiconductor characterization techniques. • to know the basics of some important spectroscopic techniques.						
Unit I	**THERMAL ANALYSIS -**Introduction – thermogravimetric analysis (TGA) – instrumentation – determination of weight loss and decomposition products – differential thermal analysis (DTA)- cooling curves - differential scanning calorimetry (DSC) – instrumentation – specific heat capacity measurements – determination of thermomechanical parameters.						
Unit II	**MICROSCOPIC METHODS -** Optical Microscopy: optical microscopy techniques – Bright field – Dark field optical microscopy – phase contrast microscopy -differential interference contrast microscopy - fluorescence microscopy - confocal microscopy - Metallurgical microscope.						
Unit III	**ELECTRON MICROSCOPY AND SCANNING PROBE MICROSCOPY-** SEM-FESEM- EDAX, - HRTEM: working principle and Instrumentation – sample preparation – scanning probe microscopy - STM – AFM - working principle, Instrumentation and modes of operation.						
Unit IV	**ELECTRICAL METHODS AND OPTICAL CHARACTERISATION-** Two probe and four probe methods- van der Pauw method – Hall probe and measurement – scattering mechanism – C-V, I-V characteristics – Schottky barrier capacitance – impurity concentration – electrochemical C-V profiling – limitations - Photoluminescence – light – matter interaction – instrumentation – Applications.						
Unit V	**SPECTROSCOPY-** Principles and instrumentation for UV-Vis-IR, FTIR spectroscopy, Raman spectroscopy, ESR, NMR, NQR, mass spectroscopy – Bain bridge-Jordan Mass spectroscope – application.						
Outcome	After completion of the course, the students shall be able to • familiar with thermal based analytical instruments applications • gain basic knowledge on various microscopes being used in scientific area for different applications • expose to recent advanced optical microscopes for high resolution surface analysis • learn use of electrical and optical methods of characterization for describing the transducer material properties • know basic and advance applications of spectroscopic techniques for molecular (simple) characterization and structure determination						

Reference and Text Books: -

1. Banwel lC. N. and McCash E.M. (2017). *Fundamentals of Molecular Spectroscopy*. McGraw-Hill Education.
2. Belk J.A. (1979). *Electron Microscopy and Microanalysis of Crystalline Materials*. Applied Science Publishers, London.
3. Kealey D.and Haines P.J. (2002). *Analytical Chemistry*. Viva Books Private Limited, New Delhi.
4. Murr L.E. (1991). *Electron and Ion microscopy and Microanalysis principles and Applications*. Marcel Dekker Inc., New York.
5. Stradling R.A. and Klipstain P.C. (1990). *Growth and Characterization of semiconductors*. Adam Hilger, Bristol.

II Semester							
CC		**Core 8**	**L**	**T**	**P**	**C**	**H/W**
Course code	**542203**	**QUANTUM MECHANICS**	**-**	**T**	**-**	**4**	**4**

Objectives	The objective of the course is to make the students • understand some important thermal analysis techniques. • familiarize with image formation in an optical microscope and learn other specialized microscopic techniques. • learn the principle of working of electron microscopes and scanning probe microscopes. • understand some important semiconductor characterization techniques. • know the basics of some important spectroscopic techniques.
Unit I	**MATHEMATICS FOR QUANTUM MECHANICS (QM) AND QM POSTULATES-** Coordinate systems, Complex numbers, Functions (odd & even, orthogonality and normalization), Differential equations, Operators: linear, differential, and Hermitian and Hamiltonian operators, Eigen functions and Eigen values, Failure of Classical Mechanics and the need for QM, Postulates of QM, time-dependent and time-independent Schrodinger wave equations
Unit II	**QM MODELS AND THEIR APPLICATIONS**–Particle in a box (1D, 2D & 3D), degeneracy and its application to linear conjugated molecular systems, free particle. Bohr's correspondence principle. QM tunnelling, Rigid Rotor: wave equation and solution calculation of rotational constants and bond length, Harmonic Oscillator: wave equation and solution, anharmonicity force constant and its significance
Unit III	**APPLICATION OF OM TO H-ATOM AND MULTI-ELECTRON ATOMS** –The Hydrogen atom and H-like ions: Solution to H and H-like wave equation, radial and angular functions, quantum numbers n, l and m and their importance. the radial distribution functions and H-like orbitals and their representation, Approximation Methods: The variation method - trial variation function and variational integral (examples of variational calculations from particle in a box)
Unit IV	**QUANTUM MECHANICAL TREATMENT OF ANGULAR MOMENTUM** – simultaneous measurement of several properties: evaluation of commutators such as [x, px), (x, px 2), (Lx, Ly] and (Lx2, Lx] and their significance. He atom: The electron spin, Pauli exclusion principle and Slater determinant for He atom. Atomic term symbols: LS and JJ coupling
Unit V	**MOLECULAR QM AND CHEMICAL BONDING** – Hydrogen molecule ion - the use of linear variation function, the LCAO method, Hydrogen molecule: Molecular orbital theory and Heitler-London treatment, electronic structure of conjugated systems: Huckel method applied to ethylene. Allyl system butadiene and benzene.
Outcome	After completion of the course, the students shall • know applications of mathematic in solving quantum chemical phenomena • understand quantum models and applications in solving the particle probability in confined mater • know how quantum mechanical theories help to identify the positioning of electrons in atoms • recognize angular momentum of particle and orbital coupling mechanism • get knowledge on quantum mechanical solutions for bonding between atoms and molecules

Reference and Text Books: -

1. A.K. Chandra, Introductory Quantum Chemistry, 4thed. Tata McGraw Hill 1994.
2. I.N. Levine, Quantum Chemistry, 5thedn, Prentice Hall, 2000.
3. P.W. Atkins, Molecular Quantum mechanics, Clarendon Press New York, 1973.
4. R.K. Prasad, Quantum Chemistry, New Age International Publishers, New Delhi, 1997.
5. R.P. Rastogi and V.K. Srivastava. An Introduction to Quantum Mechanics of Chemical Systems. Oxford &: IBH Publishing Co. New Delhi_ 1986
6. Donald A. McQuarrie, Quantum Chemistry. Viva Books PW. Ltd. New Delhi. 2003
7. P.W. Atkins and Julio de Paula. Atkins' Physical Chemistry, VII ed. Oxford University Press. 2002.

II Semester							
CC		**Core- 9**	**L**	**T**	**P**	**C**	**H/W**
Course code	**542204**	**CRYSTAL GROWTH**	**-**	**T**	**-**	**4**	**4**
Objectives	The objective of the course is to make the students • understand the concepts of nucleation and types of nucleation. • learn the theoretical aspects of crystal growth. • Learn various methods of melt growth. • acquire knowledge on the growth of crystals by solution growth. • understand various methods of growing crystals from vapour phase.						
Unit I	**NUCLEATION -**Nucleation concept – Homogeneous, heterogeneous nucleation – classical theory – Energy of formation of nucleus – kinetic theory of nucleation – statistical theory of nucleation – nucleation rate – induction period.						
Unit II	**THEORIES OF CRYSTAL GROWTH -** Two dimensional nucleation theory – Temkins model of crystal growth – limitations ofTemkins model – BCF surface diffusion theory – solution of BCF surface diffusion equation. Atmospheric nucleation.						
Unit III	**MELT GROWTH –** Temperature measurement and control – Starting materials and purification – conservative and non-conservative process – Bridgman method – Czochralski method – Verneuil method – Zone melting – Fluid flow analysis in melt growth – theory and experiment.						
Unit IV	**SOLUTION GROWTH** - Measurement of supersaturation – Low temperature solution growth – High temperature solution growth – Accelerated crucible rotation technique (ACRT) – Electrocrystallization – Crystal growth in gel – Growth of biological crystals – Hydrothermal technique – Sol-gel growth						
Unit V	**VAPOUR GROWTH -**Physical vapour transport –chemical vapor transport. Epitaxial growth techniques – Liquid phase epitaxy - vapour phase epitaxy: chloride, hydride, metalorganic - molecular beam epitaxy - chemical beam epitaxy.						
Outcome	After completion of the course, the students shall become • familiar with the concepts of nucleation and its theory. • knowledable about theories of crystal growth. • familiar with what is melt growth and various methods of melt growth. • Expert in the solution growth method. • gain knowledge on growth of crystals from its vapor phase.						

Reference and Text Books: -

1. Bhat H.L. (2019). *Introduction to crystal growth*. Tailor and Francis.
2. Brice J.C. (1986). *Crystal Growth Processes*. John Wiley and sons, New York.
3. Dryburgh P.M, Cockayane Band Barraclough K.G. (1986). *Advance Crystal Growth*. Prentice Hall, London.
4. OharaM. And Reid R.C. (1973). *Modelling Crystal Growth Rates from Solution*.
5. Pamplin B.R. (1975). *Crystal Growth*. Pergamon press, London.
6. Zxettlemoyer A.C. (1969). *Nucleation*. Marcel-Dekker Publishers.

II Semester							
CC		**Core 10**	**L**	**T**	**P**	**C**	**H/W**
Course code	**542205**	**LAB II : Materials synthesis and characterization**	**-**	**-**	**P**	**4**	**8**
Objectives	The objective of the course is to make the students • gain practical knowledge on preparation of polymers and properties • understand the physical and chemical processes of polymers • familiarize with polymer characterization methods • familiarize with use of polymer materials in different application • develop the ability to design a hypothesis driven research project and design and execute experiments to evaluate the stated hypothesis						
	1. High temperature synthesis of materials 2. Sol-gel synthesis of metal oxide composites 3. Hydrogel synthesis of oxide composites 4. Electrochemical characterization of material behaviour 5. Synthesis of nanocarbon materials 6. FTIR characterization of materials 7. TG-DTA studies of nanomaterials 8. Powder X-ray diffraction- indexing, lattice parameters and unit cell volume calculation 9. Composite of metal oxides preparation 10.Composites of carbon and metal 11.Composites of polymer and carbon 12.Composites of polymer and metal oxide 13. Alloy formation and confirmation 14. Electrochemical sensing of glucose 15. Electrochemical preparation of thin films 16. Thin film formation on metal surfaces 17. Differential pulse voltammetry technique						
Outcome	After completion of this course, students shall be able synthesize different types of nanomaterials including metal oxides, carbon nanostructures, composites and polymers. Students shall also be able to perform preliminary characterization for confirming the formation of the target material either as single phase or as composites.						

III SEMESTER

III Semester							
CC		**Core 11**	**L**	**T**	**P**	**C**	**H/W**
Course code	**542301**	**NANOMATERIALS**	**-**	**T**	**-**	**4**	**4**
Objectives	The objective of the course is to make the students • know the fundamentals of nanoscience and nanotechnology and their applications. • study the synthesis of carbon nanostructures and the mechanism of formation • know the concepts behind 1 D nanowires and nanofibers. • learn characterization of nanomaterials with various techniques. • gainknowledge of nanodevices for magnetic storage.						
Unit I	**BASIC PROPERTIES OF NANOPARTICLES -**Size effect and properties of nanoparticles - particle size - particle shape - melting point, surfacetension, wettability - specific surface area and pore size – Reason for change in optical properties, electrical properties, and mechanical properties – advantages.						
Unit II	**NANOTUBES -** Single walled and Multi walled Nanotubes (SWNT and MWNT) - synthesis and purification - synthesis of carbon nanotubes by pyrolysis techniques - arc-discharge method – CVD - nanotube properties – Nanowires – methods of preparation of nanowires –VLS mechanism.						
Unit III	**NANOWIRES AND NANOFIBERS** – Semiconductor and oxide nanowires – preparation –solvothermal – electrochemical –PVD –Pulse laser deposition – template method (qualitative)- nanofibers –electro spinning technique.						
Unit IV	**CHARACTERIZATION -** FESEM - near-field Scanning Optical Microscopy - High-resolution Transmission Electron Microscopy (HRTEM)- Absorption and emission spectra – PL spectrum - single nanoparticle characterization –Scanning capacitance microscopy – capillary electrophoresis- laser induced fluorescence (CE-LIF).						
Unit V	**NANODEVICES -** Magnetic storage: - magnetic quantum well; magnetic dots - magnetic date storage - high density quantized magnetic disks - magnetic super lattices – MRAMS - MTJs using nanoscale tunneling junctions - Millipede for storage – nano-material sensors.						
Outcome	After completion of the course, the students shall be able to • understand physical and chemical properties of nanomaterials • understand synthesis process and properties of molecular nanotubes • understand preparation methods and properties of nanowires and fibres • understand use of microscopic techniques for nanomaterial characterization • understand applications of nanomaterials in electronic and storage devices						

Reference and Text Books: -

1. EbbesenT.W. (Editor). (1997). *Carbon nanotubes: preparation and properties*. CRC Press, USA.
2. Edelstein A.S. (Editor). (1996). *Nanomaterials Synthesis, properties and applications*. IOP Publishing, UK.
3. Hari Singh Nalwa (Editor). (2000). *Hand book of Nanostructured Materials and Technology*, Vol.1-5. Academic Press, USA.
4. Hari Singh Nalwa (Editor). (2002). *Nanostructured materials and nanotechnology*. Academic Press, USA.
5. Masuo Hosokawa, Kiyoshi Nogi, Makio Naito and Toyokazu Yokoyama. (2007). *Nanoparticle Technology Handbook*. Elsevier Publishers.
6. Zhon Ling Wang. (2000). *Characterization of nanophase materials*. Wiley-VCH Verlag GmbH.

III Semester							
CC		**Core 12**	**L**	**T**	**P**	**C**	**H/W**
Course code	**542302**	**POLYMER AND COMPOSITE MATERIALS**	**-**	**T**	**-**	**4**	**4**
Objectives	The objective of the course is to make the students • learn the fundamentals of polymers, their synthesis and polymerization techniques. • gain knowledge on the various properties of polymers. • gain knowledge on various polymer processing techniques, and applications. • learn the fundamentals of composites and their mechanical behaviour. • gain knowledge on the fabrication of different types of composites.						
Unit I	**INTRODUCTION TO POLYMERS-** Classification of polymers – copolymers – tacticity – geometric isomerism – molecular weight distribution and averages — synthesis of polymers – step growth polymerisation – chain growth polymerisation – polymerisation techniques.						
UnitII	**PROPERTIES OF POLYMERS-** Polymer conformation and chain dimensions – Freely jointed chains- amorphous state – glass transition temperature – the crystalline state – ordering of polymer chains – crystalline melting temperature – techniques to determine crystallinity – Mechanical properties						
Unit III	**POLYMER PROCESSING, RHEOLOGY AND APPLICATIONS-** Basic processing operations – extrusion, molding, calendaring, coating – Introduction to polymer rheology – non-Newtonian flow – analysis of simple flows – rheometry – capillary rheometer, Couette rheometer and plate rheometer - applications-conducting polymers-biopolymers-liquid crystal polymers - high temperature polymers.						
Unit IV	**INTRODUCTION TO COMPOSITES -** Classification of composite materials – the concept of load transfer - matrix materials - polymers, metals and ceramics - fibers - glass, carbon and metallic fibers - - bonding mechanisms – mechanical behavior of composites.						
Unit V	**FABRICATION OF COMPOSITES-** Polymer matrix composites – liquid resin impregnation routes, pressurized consolidation of resin pre-pregs, consolidation of resin moulding compounds, – metal composites – squeeze infiltration, stir casting, powder blending – ceramic composites – powder-based routes, layered ceramic composites, carbon/carbon composites.						
Outcome	After completion of the course, the students shall • be familiar with different types of polymer nanomaterials • understand physical and chemical properties of polymers • be familiar with the processing of polymeric materials • understand physical and chemical properties of polymer matrix composites • acquire knowledge on polymer composite fabrication						

Reference and Text Books: -

1. Crawford R.J. (2014). *Plastics Engineering.* Elsevier India.
2. Gowarikar V.R, Viswanathan N.V.& Sreedhar J. (2019). *Polymer Science*. New Age International.
3. Hull D. & Clyne T.W. (2008). *An Introduction to Composite Materials*. Cambridge University Press.
4. Joel R. Fried. (2014). *Polymer Science and Technology*. Pearson Prentice Hall.
5. Mallick P.K. (2008). *Fiber-Reinforced Composites: Materials*, Manufacturing and Design. CRC Press, Boca Raton.

III Semester							
CC		**Core 13**	**L**	**T**	**P**	**C**	**H/W**
Course code	**542303**	**SOLID STATE PHYSICS**	**-**	**T**	**-**	**4**	**4**

Objectives	The objective of the course is to make the students • understand basic crystal structures, bonding of solids and the lattice energy calculations. • aware of electrical and thermal conduction & Fermi distribution function. • learn how our understanding of lattice dynamics is formulated in terms of travelling waves, together with the role of the interatomic forces. • Acquire adequate theoretical knowledge about solid state systems • learn the properties of semiconductors, superconductors and their applications.
Unit I	**CRYSTAL STRUCTURE AND BONDING -** Crystalline solids - crystal systems - Bravais lattices –coordination number – packing factors – cubic, hexagonal, diamond structure, Sodium Chloride Structure – lattice planes and Miller Indices– interplanar spacing – directions. Types of bonding - lattice energy - Madelung constants – Born Haber cycle – cohesive energy.
Unit II	**FREE ELECTRON THEORY -** Drude theory – Wiedemann-Franz Law and Lorentz number –Quantum state and degeneracy-density of states, concentration - free electron statistics (Fermi-Dirac), Fermi energy and electronic Specific heat, Electrical resistivity and conductivity of metals – Boltzmann transport theory –-electrical and thermal conductivity of electrons.
Unit III	**LATTICE DYNAMICS -** Mono atomic and diatomic lattices – anharmonicity and thermal expansion- phonon –Momentum of phonons, Inelastic scattering of photons by long wavelength phonons, Local phonon model – Einstein and Debye model, density of states, Thermal conductivity of solids- due to electron-due to phonons – thermal resistance of solids – phonon-phonon interaction-normal and Umklapp processes - scattering experiments.
Unit IV	**PERIODIC POTENTIALS AND ENERGY BANDS -** Bloch's theorem – Kronig-Penney model-Construction of Brillouin Zones- Effective mass of electron-nearly free electron model – Tight binding approximation-Construction of Fermi Surfaces-density of states curve-electron, holes and open orbits-Fermi surface studies - Cyclotron resonance – anomalous skin effect –de Hass van Alphen effect.
Unit V	**PHYSICS OF SEMICONDUCTORS AND SUPERCONDUCTIVITY -** Semiconductors – direct and indirect gaps – carrier statistics (intrinsic and extrinsic) – law of mass action– electrical conductivity and its temperature variation - III - V and II – VI compound semiconductors. Superconductivity – critical parameters – anomalous characteristics – isotope effect, Meissner effect – type I and II superconductors - BCS theory (elementary) - Josephson junctions and tunneling – SQUID - High temperature superconductors - applications.
Outcome	After completion of the course, the students shall be able to • understand basics on crystal structure and bonding of nanomaterials • get theoretical knowledge on movement of electrons in solids • understand theoretical knowledge on thermal energy transformation in solids • acquire knowledge on deriving energy diagrams for different types of conductors • understand physics principles of selecting semiconductors and superconductors

Reference and Text Books: -

1. Ali Omar M. (2002). *Elementary Solid State Physics*. Pearson Education.
2. Ashcroft N.Wand Mermin N.D. (2003). *Solid State Physics*, Cengage Learning.
3. Dekker A.J. (2008). *Solid State Physics*. Laxmi Publications.
4. James D. Patterson and Bernad C. Bailey. (2018). *Solid State Physics: Introduction to the Theory*. Springer. Rogalski M.S. and Palmer S.B. (2000). *Solid State Physics*. Gordon Breach Science Publishers. Wahab M.A. (2019). *Solid State Physics: Structure and Properties of Materials*. Narosa Publishing House Pvt. Ltd.

III Semester							
CC		**Core 14**	**L**	**T**	**P**	**C**	**H/W**
Course code	**542304**	**CERAMIC MATERIALS**	**-**	**T**	**-**	**4**	**4**
Objectives	The objective of the course is to make the students • learn about various processing techniques used for ceramic materials. • learn about structural ceramics and familiarize them with their properties. • get knowledge on electronic and magnetic ceramics, superconductors and fuel cells. • learn about various types of refractories. • Be aware of glass forming processes, types of glasses and their applications.						
Unit I	**CERAMIC PROCESSING-**Powder processing – precipitation, spray drying, freeze drying, sol-gel, CVD – milling techniques – forming – die pressing, slip casting, injection moulding, doctor blade processing – sintering techniques – standard pressure sintering, hot pressing, HIP, reaction bonded sintering, microwave sintering – surface finishing techniques.						
Unit II	**STRUCTURAL CERAMICS -** Oxide ceramics – zirconia, alumina, silica, mullite, magnesia and titania – carbides – silicon carbide, boron carbide, tungsten carbide, titanium carbide – nitrides – silicon nitride, boron nitride, titanium nitride, borides, silicides, - sialon – bio ceramics.						
Unit III	**ELECTRONIC CERAMICS**– Ceramic insulators and capacitors – ferroelectric ceramics – barium titanate, PZT, PLZT materials– properties and applications of electronic ceramics - magnetic ceramics – spinel ferrites, zinc ferrites – applications - garnets – superconducting ceramics – varistors – oxides and non-oxide varistors and fuel cells.						
Unit IV	**REFRACTORY CERAMICS -** Refractories – types of refractories - special refractories - silica, alumina, mullite, zirconia, cordierite - carbide based and nitride-based refractories – Fusion cast refractories – ceramic fibers– high temperature applications.						
Unit V	**GLASS CERAMICS -** Glass forming processes – Glass transition – Glass transformation range - Heat treatment schedule, crystal nucleation in glass, nucleation agent – high purity silica glass, laser glasses, fiber glasses, optical glasses and non-oxide glasses.						
Outcome	After completion of the course, the students shall • understand different processing methods ceramic • gain theoretical knowledge on structure of ceramics • understand electrical properties and characterization of ceramic materials • gain knowledge on different Refractories of ceramics • get knowledge on applications of glass ceramics in different applications						

Reference and Text Books: -

1. Cable M. And Parker J.M. (1992). *High Performance Glasses*. Chapman and Hall, London.
2. Chester J.H. (1992). *Refractories, Production and Properties*. Iron and Steel Institute, London.
3. Lewis M.H. (2011). *Glasses and Glass Ceramics*. Springer.
4. Reed J.S. (2008). *Principles of Ceramic Processing*. Wiley-Inter science.
5. Richerson D.W. & Lee W.E. (2018). *Modern Ceramic Engineering: Properties, Processing and Use in design*. CRC Press.

III Semester							
CC		**Core 15**	**L**	**T**	**P**	**C**	**H/W**
Course code	**542305**	**LAB III:** Device Characterization	**-**	**-**	**P**	**4**	**8**
Objectives	The objective of the course is to equip the students with adequate experimental skill which includes design and testing of electronic components, devices and sensors. Students shall also perform thermal and mechanical properties of select materials						
Practical	1. Characteristics of PN junction diode 2. Zener diode Characteristics Regulator using Zener diode 3. Common Emitter input-output Characteristics 4. Common Base input-output Characteristics 5. FET and SCR Characteristics 6. Clipper and clamper circuits 7. Verifications of Thevenin & Norton theorem 8. Verifications of Kirchhoff's Voltage Law and Kirchhoff's Current Law 9. Verifications of Super position theorem 10. Verifications of maximum power transfer & reciprocity theorem 11. Determination of resonance frequency of series & parallel RLC circuits 12. Transient analysis of RL and RC circuits 13. Resistive sensors 14. Stress and strain study (tensile strength, compression & bending moment) 15. Thermal properties of materials 16. Impact test and surface roughness study 17. Corrosion studies						
Outcome	At the end of the course work, students shall have gained adequate training and skills in the design and testing of electronic components, devices and sensors. In addition, they also become familiar about testing mechanics and thermal properties of select materials.						

IV SEMESTER

IV Semester							
		Core 16	**L**	**T**	**P**	**C**	**H/W**
Course code	**542999**	**Project/Internship**	**-**	**-**	**P**	**15**	**30**

Objectives

The overall aim of the research project/internship is

- to allow students to demonstrate the personal abilities and skills required to produce and present an extended piece of work.
- To engage in personal inquiry, action and reflection on specific topics and issues.
- To focus on, and demonstrate an understanding of, the areas of interaction.
- To reflect on learning and share knowledge, views and opinions.

Specific objectives for allowing one full semester to carry out research project and/or to undergo are internship training programme are as follows;

A) Planning and Development

Students should:

- identify a clear and achievable goal
- describe and justify a focus on the chosen area(s) of interaction
- describe the steps followed to achieve the stated goal
- adhere to the stated goal throughout the project.

B) Literature/data collection

Students should:

- select and utilize adequate, varied resources
- identify and use relevant information critically
- acknowledge sources of information appropriately.

C) Choice and Application of Techniques

Students should:

- choose experiment/techniques relevant to the project's goal
- justify this selection
- apply the chosen techniques consistently and effectively.
- Acquire sufficient data relevant to the goal

D) Data analysis

Students should:

- analyse the data in terms of the goal and the focus of the project
- express personal thought
- support arguments with evidence
- respond thoughtfully to ideas and inspiration.

E) Organization of the Dissertation

Students should:

- organize their work in a coherent manner according to the required structure
- present information clearly
- present references, bibliography and symbolic representations appropriately.

F) Analysis of the Process and Outcome

Students should:

- identify the strengths and weaknesses of the project at different stages of development
- where appropriate, suggest ways in which the project could have been tackled differently
- assess the achieved results in terms of the initial goal and the focus on the chosen area(s) of interaction
- show awareness of the overall perspectives related to the chosen topic or piece of work.

	G) Personal Engagement Students should: • meet deadlines • follow agreed procedures and work plans • make appropriate use of a process journal or log book • show initiative, enthusiasm and commitment to the task.
Outcome	After carrying the project/internship as detailed above, the students get to know the ways and means to do a research project on the chosen topic. By doing such intensive project work, students become ready for job in any industry or institution. Depending on the quality of the project work and the students involvement in doing the sane, students most get placed either in the campus or outside campus.

ELECTIVE PAPERS
DISCIPLINE SPECIFIC ELECTIVE

I Semester							
DSE 1		**Elective**	**L**	**T**	**P**	**C**	**H/W**
Course code	**542501**	**BIOMATERIALS**	**-**	**T**	**-**	**3**	**3**

Objectives	The objective of the course is to make the students • learn the response of biomaterials to host environment, and host response to biomaterials. • gain knowledge on various materials used in bone and joint replacement. • study about materials used in cardiovascular implants. • know about dental materials and dental implants. • gain knowledge on soft tissue and drug delivery materials.
Unit I	**BIOLOGICAL PERFORMANCE OF MATERIALS -**Biocompatibility- introduction to the biological environment – material response: swelling and leaching, corrosion and dissolution, deformation and failure, friction and wear – host response: the inflammatory process - coagulation and hemolysis- approaches to thrombo- resistant materials development.
Unit II	**ORTHOPAEDIC MATERIALS -** Bone composition and properties - temporary fixation devices - joint replacement – biomaterials used in bone and joint replacement: metals and alloys – stainless steel, cobalt-based alloys, titanium-based materials – ceramics: carbon, alumina, zirconia, bioactive calcium phosphates, bioglass and glass ceramics – polymers: PMMA, UHMWPE/HDPE, PTFE – bone cement – composites.
Unit III	**CARDIO VASCULAR MATERIALS**- Blood clotting – blood rheology – blood vessels – the heart – aorta and valves – geometry of blood circulation – the lungs - vascular implants: vascular graft, cardiac valve prostheses, cardiac pacemakers – blood substitutes – extracorporeal blood circulation devices.
Unit IV	**DENTAL MATERIALS**- Teeth composition and mechanical properties – impression materials – bases, liners and varnishes for cavities – fillings and restoration materials – materials for oral and maxillofacial surgery – dental cements and dental amalgams – dental adhesives.
Unit V	**SOFT TISSUE MATERIALS -** Biomaterials in ophthalmology – viscoelastic solutions, contact lenses, intraocular lens materials – tissue grafts – skin grafts – connective tissue grafts - suture materials – tissue adhesives – drug delivery: methods and materials – selection, performance and adhesion of polymeric encapsulants for implantable sensors.
Outcome	After completion of the course, the students shall • become knowledgeable on the biocompatibility and its response to the host. • gain knowledge on the application of materials in bone and joint replacement. • know about the application of materials in cardio vascular implants. • Become familiar with how the materials are used for the purpose of dental implants. • Be aware of application of biomaterials in ophthalmology and implantable sensors.

Reference and Text Books: -

1. Black J. (1992). *Biological Performance of Materials: Fundamentals of Biocompatibility*. Marcel Dekker Inc, New York.
2. Chen Q.andThouas G. (2015). *Biomaterials. A Basic Introduction*. CRC Press.
3. Park J.& Lakes R.S. (2010).*Biomaterials: An Introduction*. Springer.
4. Ratner B.D, Hoffman A.S, Schoen F.J.& Lemons J.E. (2004). *Biomaterials Science: An Introductionto Materials in Medicine*. Academic Press.
5. Sujata V. Bhat. (2019). *Biomaterials*. Springer.
6. Williams D.F. (editor). (1992).*Materials Science and Technology: A Comprehensive treatment*, Volume 14. Medical and Dental Materials. VCH Publishers Inc, New York.

I Semester							
DSE 1		**Elective**	**L**	**T**	**P**	**C**	**H/W**
Course code	**542502**	**MOLECULAR ELECTRONICS**	**-**	**T**	**-**	**3**	**3**

Objective	The objective of the course is to make the students • know the basics of the role of molecules in electronics industry. • gain knowledge on carbon nanostructures • learn about fundamentals of nano electronic devices. • familiar with molecular computers and its application. • learn about plastic electronics and optoelectronic devices.
Unit I	**INTRODUCTION TO NANOTECHNOLOGY -**Background to nanotechnology: periodic table – atomic structure – molecules and phases – energy – molecular and atomic size – surface and dimensional space – Nanaomaterials-top down and bottom up approaches.
UnitII	**CARBON NANOSTRUCTURES -** Fullerenes – CNTs-types of nanotubes – formation of nanotubes – assemblies – purification of carbon nanotubes – electronic properties – synthesis of carbon nanotubes – carbon nanotube interconnects – carbon nanotube FETs – CNTs for memory applications.
Unit III	**NANOELECTRONIC DEVICES-FUNDAMENTALS** - Electrodes & contacts – functions – molecular electronic devices – first test systems – simulation and circuit design – fabrication; Future applications: MEMS – robots – random access memory – mass storage devices.
Unit IV	**MOLECULAR COMPUTERS** - Molecular wires and switches. Biomolecular computer, molecular arrays as memory stores, DNA for molecular devices - DNA's ability to conduct electrical currents. Charge transfer rates in solution - molecules between nanofabricated electrodes.
Unit V	**FLEXIBLE ELECTRONIC DEVICES -** Electroactive organic molecules, Plastic Electronics, Electrical conduction in polymers, Donor molecules, Acceptor molecules, Optoelectronic devices: OLEDs, OTFTs.
Outcome	After completion of the course, the students shall be able to • get basic knowledge about nanotechnology. • know about different carbon materials and their applications. • get familiar with fundamentals of nano-electronic devices and MEMS. • get knowledge on molecular computer and DNA molecular devices. • understand the application of flexible devices in optoelectronics.

Reference and Text Books: -

1. Aswal D. K., Yakhmi J.V. (2010). *Molecular and Organic Electronics Devices (Electrical Engineering Developments) Ed*. Nova Science Pub Inc; 1 edition.
2. Geoffrey J, Ashwell. (1992). *Molecular Electronics*, Ed., John Wiley & Sons Inc.
3. James M Tour. *Molecular Electronics; Commercial Insights, Chemistry, Devices, Architecture and Programming*: (Rice University, USA), ISBN: 978-981-238-269.
4. Juan Carlos Cuevas and Elke Scheer. (2010). *Molecular Electronics: An Introduction to Theory and Experiment (Nanotechnology and Nanoscience) (World Scientific Series in Nanotechnology and Nanoscience)*. World Scientific Publishing Company; 1st Edition.
5. Michael C. Petty. *Molecular Electronics: From Principles to Practice (Wiley Series in Materials for Electronic & Optoelectronic Applications).*
6. Michael Wilson, Kamali Kannangara, Geoff Smith, Michelle Simmons and Burkhard Q Raguse.(2002). *Nanotechnology: Basic Science and Emerging Technologies*, Chapman & Hall CRC.
7. Pradeep T. (2007). *NANO: The Essentials – Understanding Nanoscience and Nanotechnology*.
8. Rainer Waser (Ed.), (2003). *Nanoelectronics and Information Technology: Advanced Electronic Materials and Novel Devices*, Wiley-VCH.

I Semester							
DSE 1		**Elective**	**L**	**T**	**P**	**C**	**H/W**
Course code	**542503**	**NON-DESTRUCTIVE TESTING**	**-**	**T**	**-**	**3**	**3**
Objectives	The objective of the course is to make the students • learn about liquid penetrant and magnetic particle inspection. • understand the principle, working and uses of radiographic testing. • gain knowledge about the ultrasonic testing. • understand the principle, working and application of eddy current technique. • understand the thermal and optical methods used in NDT.						
Unit I	**INTRODUCTION AND SURFACE NDT METHODS -**Definition of terms, discontinuities and defects/flaws – fracture mechanics concept of design and the role of NDT – life extension and life prediction – penetrant testing and magnetic particle testing, basic principle of penetrant testing – limitations and advantages – basic principle involved in magnetic particle testing – development and detection of large flux – longitudinal and circular magnetization – demagnetization.						
Unit II	**RADIOGRAPHIC TESTING -** Properties of X-rays and gamma rays – attenuation and differential attenuation – interaction of radiation with matter – Principle of radiographic testing and recording medium – films and fluorescent screens – nonimaging detectors – film radiography – calculation of exposure for X-ray and gamma rays – quality factors – Image quality indications and their use in radiography – neutron radiography.						
Unit III	**ULTRASONIC TESTING** - Ultrasonic waves – velocity, period, frequency and wavelength – reflection and transmission – near and far field effects and attenuation – generation – piezoelectric and magnetostriction methods – normal and angle probes – methods of Ultrasonic testing – Principle of pulse echo method – Equipment – examples – rail road inspection, wall thickness measurement – range and choice of frequency.						
Unit IV	**EDDY CURRENT TESTING** - Introduction – Principles of eddy current inspection – conductivity of a material – magnetic properties – coil impedance – lift off factor and edge effects – skin effect – inspection frequency – coil arrangements – inspection probes – types of circuit – Reference pieces – phase analysis – display methods – typical applications of eddy current techniques.						
Unit V	**THERMAL AND OPTICAL METHODS-** Imaging – principle and applications – testing of composites – acoustic emission testing – application of AET – on-line monitoring or continuous surveillance and applications in materials science – Optical methods of NDT – photo elasticity – evaluation procedure – Holographic NDT procedure – speckle phenomenon – speckle interferometry – speckle shear interferometry – Fourier optics – Fourier filtering techniques for non-destructive testing.						
Outcome	After completion of the course, the students shall be able to • get familiar with various testing methods of magnetic particles and their advantages. • understand the role of electromagnetic rays in testing. • know how ultrasonic waves used in testing of materials. • get knowledge on testing of materials using eddy current. • get familiar with application of thermal and optical methods in testing of composites.						

Reference and Text Books: -

1. Dainty J.C. (1984). *Laser Speckle & Related Phenomena*, Springer-Verlag, New York.
2. Hull B. and John V. (2012). *Non-Destructive Testing*. Springer-Verlag New York Inc.
3. McGonnagle W.J. (1961). *Non-Destructive Testing Methods*, Mc Graw Hill Co., NY.
4. *Metals Hand Book*, Vol.2, 8th Edition, ASTM, Metals Park, Ohio.

I Semester							
DSE 1		**Elective**	**L**	**T**	**P**	**C**	**H/W**
Course code	**542504**	**NONLINEAR OPTICS AND MATERIALS**	**-**	**T**	**-**	**3**	**3**
Objectives	The objective of the course is to make the students • understand the concepts of electromagnetic theory and refractive index of materials. • Read the concept of optical susceptibility. • understand the concept of second order non linearity. • get knowledge on the processes of third order nonlinear optical effects. • understand the properties of non-linear optical materials.						
Unit I	**ELECTROMAGNETIC THEORY -**Maxwell equations – wave equations in various media and its propagation – origin of complex refractive index – classical theory of optical absorption (electron oscillator model) and dispersion (Lorenz oscillator model) – classical theory ofanharmonic oscillators.						
Unit II	**OPTICAL SUSCEPTIBILITIES -** Wave equation description of nonlinear optical susceptibilities – quantum mechanical treatment of nonlinear optical susceptibilities – frequency and intensity dependence of polarization – and dielectric susceptibility – first and higher order susceptibilities.						
Unit III	**SECOND-ORDER NONLINEARITIES** - Second harmonic generation – sum and difference frequency generation – parametric processes – simple theory and calculations of nonlinear polarization – various phase matching techniques in second harmonic generation (SHG).						
Unit IV	**THIRD-ORDER NONLINEARITIES** - Third harmonic generation – four-wave mixing – Kerr nonlinearity – intensity dependent effect – self-phase modulation – cross-phase modulation. Stimulated Raman scattering – stimulated Brillioun scattering. Parametric gain – parametric amplification and oscillation -. Applications of frequency mixing and up-conversion – difference frequency generation – optical phase conjugation: theory and applications – Photorefractive effect and applications – solitons: theory and applications – optical bistability.						
Unit V	**NONLINEAR OPTICAL MATERIALS -** Nonlinear optics of organic materials and polymers – liquid crystals – photorefractive materials – organic doped glasses – rare earth doped glasses and crystals – semiconductors – optical fibers and photonic crystal fibers – ferroelectric materials and other novel optical materials.						
Outcome	After completion of the course, the students shall be able to • understand the concept of electromagnetic theory. • get familiar with optical susceptibility and order of susceptibility. • know the origin of second harmonic generation and techniques in generation of second harmonic generation. • understand the important third order nonlinearities. • gain knowledge on nonlinear optical materials.						

Reference and Text Books: -

1. Bloembergen N. (2005). *Nonlinear Optics*. World Scientific, Singapore.
2. LaudB.B. (2011). *Lasers and Non-linear Optics*. New Age International Pvt. Ltd.
3. MurtiY.V.G.S. andVijayanC. (2014). *Essentials of Nonlinear Optics*. Wiley.
4. Robert W.Boyd. (2009). *Nonlinear Optics*. Academic Press, London.
5. Shen Y.R. (2003). *Principles of Nonlinear optics*. Wiley-Interscience, New York.
6. Singh N.B. (1990). *Growth and characterization of nonlinear optical materials*. Pergamon Press.

I Semester							
DSE 1		**Elective**	**L**	**T**	**P**	**C**	**H/W**
Course code	**542505**	**LASERS AND APPLICATIONS**	**-**	**T**	**-**	**3**	**3**
Objectives	The objective of the course is to make the students • understand about theoretical studies on laser systems. • get basic knowledge on laser system compound. • get knowledge about various laser systems. • able to know about laser system used for materials processing. • get knowledge on the laser applications.						
Unit I	**PRINCIPLES OF LASERS-**Spontaneous emission, Stimulated emission, Einstein coefficients, ratio of rates of stimulated and spontaneous emission – Threshold condition for laser action – Rate equations – Population inversion in three level and four level systems.						
Unit II	**OPTICAL RESONATORS-** Resonant cavities, Gaussian beam characteristics, resonator modes, spot size – Types of resonators, geometries, quality factor of an optical resonator – Q-switching and Modelocking concepts and techniques.						
Unit III	**LASER SYSTEMS**- Gas lasers: He-Ne laser, Carbon dioxide gas laser, Nitrogen gas laser, Argon ion gas laser – Solid state lasers: Ruby laser, Nd-YAG laser, fiber laser, Ti-Sapphire - Semiconductor Laser – homojunctionand heterojunction laser - Liquid Lasers: Dye lasers.						
Unit IV	**MATERIALS PROCESSING**- Laser power density – heat affected zone - Welding - Fusion depth and welding geometry - Welding speeds - Advantages and uses of laser welding - Drilling hole geometry - Advantages and uses of laser drilling - resistor trimming - Capacitor height adjustment and fabrication, Scribing - Controlled fracturing.						
Unit V	**APPLICATIONS -** Metrology – inter-ferrometric techniques - Laser ranging and tracking - Laser Doppler velocimetry - Ring laser and rotation sensing - Pollution monitoring - Holography and speckle in displacement and deformation measurements – ions – Medical applications.						
Outcome	After completion of the course, the students shall be able to • get familiar with principles of laser action. • get knowledge on Q-switching and mode locking. • know about the working methods of various laser methods. • gain knowledge on various laser processing methods. • get familiar with application of lasers in various fields.						

Reference and Text Books: -

1. Charchan S.S. (1975). *Lasers in Industry*. Van Nostrand Reinhold Co.
2. Laud B.B. (2011). *Lasers and Non-Linear Optics*. New Age International (P) Ltd.
3. Shea D.C.O, Callen W.R and Rhodes W.T. (1977). *An Introduction to Lasers and their Applications*. Pearson.
4. Steen William M. (2010). *Laser Material Processing*. Springer.
5. Verdeyen J.T.(1990).*Laser Electronics*. Prentice Hall.

I Semester							
DSE 1		**Elective**	**L**	**T**	**P**	**C**	**H/W**
Course code	**542506**	**PYTHON PROGRAMMING**	**-**	**T**	**-**	**3**	**3**
Objectives	The objective of the course is to make the students • know the concepts of algorithms and developing them. • understand different types of data expressions and statements in Python environment. • know the aspects of control flow and functions in Python environment. • know the concepts of lists, tuples and dictionaries in Python environment. • learn to use files, modules and packages.						
Unit I	**ALGORITHMIC PROBLEM SOLVING -**Algorithms, building blocks of algorithms (statements, state, control flow, functions), notation (pseudo code, flow chart, programming language), algorithmic problem solving, simple strategies for developing algorithms (iteration, recursion). Illustrative problems: find minimum in a list, insert a card in a list of sorted cards, guess an integer number in a range, Towers of Hanoi.						
Unit II	**DATA, EXPRESSIONS, STATEMENTS -** Python interpreter and interactive mode; values and types: int, float, boolean, string, and list; variables, expressions, statements, tuple assignment, precedence of operators, comments; modules and functions, function definition and use, flow of execution, parameters and arguments; Illustrative programs: exchange the values of two variables, circulate the values of n variables, distance between two points.						
Unit III	**CONTROL FLOW, FUNCTIONS –** Conditionals: Boolean values and operators, conditional (if), alternative (if-else), chained conditional (if-elif-else); Iteration: state, while, for, break, continue, pass; Fruitful functions: return values, parameters, local and global scope, function composition, recursion; Strings: string slices, immutability, string functions and methods, string module; Lists as arrays. Illustrative programs: square root, gcd, exponentiation, sum an array of numbers, linear search, binary search.						
Unit IV	**LISTS, TUPLES, DICTIONARIES** - Lists: list operations, list slices, list methods, list loop, mutability, aliasing, cloning lists, list parameters; Tuples: tuple assignment, tuple as return value; Dictionaries: operations and methods; advanced list processing - list comprehension; Illustrative programs: selection sort, insertion sort, mergesort, histogram.						
Unit V	**FILES, MODULES, PACKAGES -**Files and exception: text files, reading and writing files, format operator; command line arguments, errorsand exceptions, handling exceptions, modules, packages; Illustrative programs: word count, copy file.						
Outcome	After completion of the course, the students shall be able to • gain the basic knowledge on algorithm. • get familiar with data expressions types and statements in Python environment. • know to use control flow and functions in python environment. • know to use lists, tuples and dictionaries in python environment. • know to use files, modules and packages in python environment.						

Reference and Text Books: -

1. Eric Matthes. (2015). *Python Crash Course*. No Starch Press.
2. Kenneth Lambert. (2012). *Fundamentals of Python: First Programms*. Cengage Learning.
3. Mark Lutz. (2013). *Learning Python*. O'Reilly Media.
4. Nageswara RaoR. (2019). *Core Python Programming*. Dreamtech Press.
5. Yuxi Liu.(2017).*Python Machine Learning by Example*. Packt Publishing Ltd.

II Semester							
DSE 2		**Elective**	**L**	**T**	**P**	**C**	**H/W**
Course code	**542507**	**Molecular spectroscopy**	**-**	**T**	**-**	**3**	**3**

Objectives	The objective of the course is to make the students • gain knowledge on various aspects of spectroscopic techniques • learn the concepts of various techniques in molecular characterization. • know about different molecular movements. • gain knowledge on Raman effect and Raman spectroscopy. • gain knowledge on resonance spectroscopy.
Unit I	**SYMMETRY ASPECTS OF MOLECULAR ORBITALS -**Valence bond theory – Molecular orbital theory- Heitler London theory for Hydrogen molecule - Hybridization – SP – SP^2&SP^3 Hybrids.
Unit II	**ROTATIONAL SPECTRA -** Rotational energy of a diatomic molecule – Rigid and non-rigid rotators – isotopic substitution – Stark effect – its importance in microwave spectroscopy – quadrupole hyperfine interaction - Rotational spectra of polyatomic molecules – pure rotational Raman spectra – diatomic linear molecule – symmetric top molecules- Molecular structure – using IR & Raman spectroscopy.
Unit III	**VIBRATIONAL PROPERTIES-** Vibrational spectra of diatomic and polyatomic molecules – Information on molecular constitution from IR studies – Vibrational Raman spectra – Vibrational course structure – Rotational course spectra – Franck – Condon principle – intensity distribution – portrait parabolae – disassociation – pre-disassociation – mutual exclusion principle.
Unit IV	**NON LINEAR SPECTROSCOPIC PHENOMINA -** Non-linear Raman Phenomena-Hyper Raman effect- Classical treatment –Experimental techniques- Stimulated Raman Scattering –Inverse Raman Effect-Coherent Anti-Stoke's Raman Scattering-Photo acoustic Raman Scattering-Multi photon spectroscopy-two photon absorption-Multiphoton absorption. X-ray spectra; rotational and vibrational spectra of diatomic molecules.
Unit V	**RESONANCE SPECTROSCOPY -** Interaction between spin and magnetic field – Nuclear resonance – Bloch equations - Chemical shift – Dipole –Dipole interaction and spin lattice interaction – Mossbauer –ESR-NQR (principle only) spectroscopy and its application – Mossbauer spectroscopy - applications – Electronic structure – molecular structure – crystal symmetry and molecular structures.
Outcome	After completion of the course, the students shall • become familiar with representation of different molecular structures • be aware of rotational movements in molecules and their identification • know rotational movements in molecules and their identification • gain detailed analytical knowledge on the utility of Raman effect • get to know electromagnetic wave resonance with molecular movement and interaction of functional groups

Reference and Text Books: -

1. Aruldhass G. (2001). *Molecular structure and spectroscopy*. Prentice Hall of India, New Delhi.
2. Colin N Banwell. (2019). *Fundamentals of Molecular Spectroscopy*, McGraw Hill.
3. Dogra S.K. (2015).*Atomic and Molecular Spectroscopy.* Pearson Publications.
4. Mchale, Jeanne L. (2008).*Molecular Spectroscopy*. Pearson Publications.
5. MoolCh and Guptha. (2001).*Atomic and Molecular Spectroscopy*. New age International Publishers, New Delhi,
6. Sindhu P.S. (2010).*Elements of Molecular Spectroscopy*. New Age International.
7. Willard et al. (2005). *Instrumental methods of analysis*. CBS publishers

II Semester							
DSE 2		**Elective**	**L**	**T**	**P**	**C**	**H/W**
Course code	**542508**	**SEMICONDUCTOR MATERIALS AND DEVICES**	**-**	**T**	**-**	**3**	**3**

Objectives	The objective of the course is to make the students • know the basic aspects of electronic energy band structures. • understand optical properties of materials. • understand transport properties of charges in materials. • learn the aspects of fabrication of semiconductor devices. • understand fabrication of optoelectronic devices.
Unit I	**INTRODUCTION -**Introduction: Properties of semiconductors –Free electron Theory - Transport properties. Bonds and Band sin Semiconductor: -Electronic and structure - Junction Properties of semiconductors- Recombination mechanism - Electron, Hole recombination through traps - Junction properties of p-n, n+-n, p+-p junctions - Surface recombination – Recombination with donors and acceptors at low temperatures - Quantum theory of junction devices - Generation of recombination processes injunction devices.
Unit II	**OPTICAL PROPERTIES -** Optical properties of semiconductors - Optical constants - Light absorption spectrum – Light absorption edge - Effect of free charge carriers on the absorption edge - Fundamentals of absorption and reflection- Electron transport phenomena: Theory of electron transport in crystalline semiconductors - Boltzmann's transport equation for Bloch states - relaxation time - relaxation time approximation to the low field transport coefficients - scattering mechanism.
Unit III	**TRANSPORT PROPERTIES –** Basic Process in Semiconductor Devices: Equilibrium properties - electrons and holes – impurities in semiconductors - carrier concentration as a function of temperature - High doping effects - Non-equilibrium phenomena - carrier transport - Transport properties in high fields – recombination and generation processes - breakdown mechanism – Basic equations for Semiconductor devices - equations for the interior of devices – boundary conditions – Systems.
Unit IV	**FABRICATION OF TRANSISTORS AND THYRISTORS** - Unipolar devices: Metal-Semiconductor contacts – Energy- Band Relation - Schottky Effect - Characterization of Barrier Height - Device Structure - Ohmic Contact - JFET and MESFET - basic device characteristic - general characteristic - MOSFET - basic device characteristic - MOSFET Structures – Non-volatile memory devices. Bipolar transistor - Static characteristics - power transistor - switching transistor – Thyristors - basic characteristics - Schottky diode - Three terminal thyristor.
Unit V	**FABRICATION OF LED'S AND SENSORS -**Photonic Devices: Light Emitting diodes - LED for fiber optics - LED performance - reliability - Photodetectors - Photoconductor - Photodiode - Avalanche Photodiode - Phototransistor - Solar cells - Thin film solar cells - solid state sensors, optical Sensors - optoelectronic components.
Outcome	After completion of the course, the students shall be able to • get knowledge on semiconductors and energy band structures. • study about optical properties of materials. • get familiar with role of impurities in semiconductor and electron transport theory. • get knowledge on fabrication of transistors and thyristors. • know different steps in device fabrications.

Reference and Text Books: -

1. Sze S.M. and NgK. K. (2008). *Physics of Semiconductor devices*. Wiley India, New Delhi.
2. Wolfe C.M, Holonyak J. R. NandStillman G.E. (1989).*Physical Properties of Semiconductors*. Prentice Hall International Inc., London.
3. Butcher P.N, March N.H. and Tosi M.P. (2014). *Crystalline Semiconducting Materials and Devices*. Springer.
4. Fraser D.A. (1986). *The Physics of Semiconductor devices*. Clarendon Press, Oxford.
5. Schroder D.K. (1990). *Semiconductor Material and Device Characterization*. JohnWiley& Sons Inc., New York.
6. Keller S.P. (1980). *Handbook on Semiconductors*, Vol. 1-4. T.S. Moss, Ed., North-Holland, Amsterdam.

II Semester							
DSE 2		**Elective**	**L**	**T**	**P**	**C**	**H/W**
Course code	**542509**	**PHYSICAL METALLURGY**	**-**	**T**	**-**	**3**	**3**
Objectives	The objective of the course is to make the students • know the concepts of phase diagrams. • get knowledge about iron carbon phase equilibrium diagram and alloys. • learnabout various heat treatment processes those are employed. • understand about various phase transformations. • get knowledge various engineering alloys and their applications.						
Unit I	**PHASE DIAGRAMS -**Composition and classification of pig iron and cast iron – iron ores - manufacture of wrought iron and steel - The phase rule - Types of Binary Diagrams,– invariant reactions- eutectic, eutectoid, peritectic and peritectoid reactions – Thermodynamics, Solution theory - free energy composition curves – Experimental determination of equilibrium diagram-grain size analysis, grain size measurement - effect of grain size on properties of metals and alloys.						
UnitII	**SOLID SOLUTION -** Types of solid solution – solid solution factors governing substitutional solubility –Hume- Rothery rules- intermediate phases -solid solution alloys –Vegards law – Lever rule - mechanical mixtures-- Iron-Carbon equilibrium diagram – Aluminum alloys – Copper alloys – Effect of alloying elements.						
Unit III	**HEAT TREATMENT** – Recovery, recrystallisation and grain growth: property changes, annealing twins, textures in cold worked and annealed alloys,-TTT diagrams – CCT diagrams – heat-treatment processes – annealing, normalising, quenching and tempering – baths used in heat treatment – hardenability – Jominy's end quench test – martempering and austempering – case hardening – induction, flame, laser - carburising, cyaniding, nitriding, carbo nitriding.						
Unit IV	**PHASE TRANSFORMATIONS -** Types of phase changes – Driving forces, N-G aspects, diffusion in solids – solidification – pearlitic transformations – martensitic transformations – kinetics of transformation - precipitation and age hardening.						
Unit V	**ENGINEERING ALLOYS -** Low carbon steels – mild steels – high strength structural steels – tool materials – stainless steels – super alloys – light alloys – shape memory alloys – applications.						
Outcome	After completion of the course, the students shall be able to • get familiar with phase diagrams and will study the effects of grain size. • get knowledge on solid solutions and alloys. • gain knowledge on various heat treatment process. • understand the mechanism and kinetics of phase transformation. • get knowledge on how to engineer alloys of interest and their applications.						

Reference and Text Books: -

1. Avner S.H. (2019). *Introduction to Physical Metallurgy*. Mc Graw Hill Education.
2. GuyA.G. and Hren J. (1984). *Elements of Physical Metallurgy*. Oxford Univ. Press.
3. Lakhtin Y. (2005). *Engineering Physical Metallurgy*. CBS Publishers & Distributors.
4. PolmearI. S. (1995). *Light Alloys*. *Metallurgy and Materials Science*.
5. Raghavan V. (2015). *Physical Metallurgy: Principles and Practice*. PHI Learning Private Limited, New Delhi.
6. Robert E. Reed-Hill. (2008). *Physical Metallurgy Principles*. Affiliated East-West Press.
7. Smith W.F. (2014). *Structural Properties of Engineering Alloys*, McGraw Hill Education

II Semester							
DSE 2		**Elective**	**L**	**T**	**P**	**C**	**H/W**
Course code	**542510**	**MATERIALS PROCESSING**	**-**	**T**	**-**	**3**	**3**
Objectives	The objective of the course is to make the students • understand the basics of manufacturing processes. • get knowledge about surface treatment processes. • know about various processes of welding. • learn about mechanical working of metals. • understand the knowledge about powder metallurgical processes.						
Unit I	**BASIC MANUFACTURING PROCESSES -**Fundamental analysis of Manufacturing processes, casting, casting processes, forging, methods of forging, extrusion, rolling, spinning, turning, planning and shaping, milling, grinding.						
Unit II	**SURFACE TREATMENT PROCESSES -** Necessity for surface modification, surface cladding, surface alloying, hard facing, shock hardening, conventional methods, carburising, nitriding, cyaniding, advantages of laser surface treatment over conventional methods, typical laser variables used in surface alloying, laser cladding, experimental set up.						
Unit III	**WELDING PROCESSES –** Various processes of welding, fusion welding, pressure welding, oxyacetelene welding, resistance welding, spot welding, thermit welding, hermetic welding, projection welding, seam welding, butt welding, thermal effects of welding, effects on grain size and microstructure, internal stresses effect, corrosion effect, high energy beam welding, laser beam and electron beam welding, key hole effect.						
Unit IV	**MECHANICAL WORKING OF METALS** - Hot working, cold working, normalising, full annealing, tempering, theory of tempering, effect of tempering temperature on mechanical properties of carbon steels, different tempering process, deformation of metals, elastic deformation, plastic deformation, slip, twinning – assessment of processed materials.						
Unit V	**POWDER METALLURGICAL PROCESS -**Production of powders, powder mixing, compacting, types of presses, sintering, soaking, finishing process, limitations and advantages of powder metallurgy, applications, production of cemented carbide cutting tools, self-lubricating bearings, magnets, cermets, ultrasonic ceramic transducers.						
Outcome	After completion of the course, the students shall be able to • get know about various manufacturing processes. • get familiar with surface treatment processes. • know the concepts of welding techniques. • know about the mechanical working of metals. • get familiar with powder metallurgical process.						

Reference and Text Books: -

1. Gupta R.B. (1995). *Materials Science and Processes*. Satya Prakashan, New Delhi.
2. Muralidhara M.K. (1998). *Materials Science and Processes*. Dhanpat Rai Publishing Co., Delhi.
3. Rajan T.V, Sharma C. P and Sharma A.(2010). *Heat Treatment: Principles and Techniques*. Prentice Hall India Learning Private Limited.
4. Rykalin, Uglov A, Kokona, (1987). *A Laser and Electron beam material processing hand book*, MIR Publishers.

II Semester							
DSE 2		**Elective**	**L**	**T**	**P**	**C**	**H/W**
Course code	**542511**	**CORROSION SCIENCE AND ENGINEERING**	**-**	**T**	**-**	**3**	**3**
Objectives	The objective of the course is to make the students • understand corrosion process and corrosion control. • understand the methods used for testing corrosion. • getknowledge on different methods used for coating. • get knowledge on various types of corrosion with respect to corrosion. • learnabout various application of coating.						
Unit I	**CORROSION PROCESSES -**Basic principles of electrochemistry and aqueous corrosion processes - Electrochemical Thermodynamics and Electrode Potential - Electrochemical Kinetics of Corrosion Cathodic and anodic behavior - Faraday's Law - Nernst equation; standard potentials Pourbaix diagram - Tafel equations, corrosion rate - Evans diagram - pitting, crevice and exfoliation corrosion; influence of deposits and anaerobic conditions; corrosion control; high temperature oxidation and hot corrosion; corrosion/mechanical property interactions.						
Unit II	**CORROSION TESTING -** Materials and specimens – surface preparation – measuring and weighing – linear polarization – AC impedance – *in vivo* corrosion – paint tests – seawater tests.						
Unit III	**COATING MANUFACTURE –** Electrode position; flame and plasma spraying; thermal, HV of detonation gun, physical vapour deposition; chemical vapour deposition; HIP surface treatments.						
Unit IV	**CORROSION IN SELECTED ENVIRONMENTS** - Atmospheric Corrosion, Corrosion in Automobiles, Corrosion in Soils, Corrosion of Steel in Concrete, Corrosion in Water, Microbiologically Induced Corrosion, Corrosion in the Body, Corrosion in the Petroleum Industry, Corrosion in the Aircraft Industry, Corrosion in the Microelectronics Industry.						
Unit V	**COATING APPLICATIONS -**Abrasive, erosive and sliding wear. The interaction between wear and corrosion. Coating systems for corrosion and wear protection; new coating concepts including multi-layer structures, functionally gradient materials, intermetallic barrier coatings and thermal barrier coatings.						
Outcome	After completion of the course, the students shall be able to • get familiar with corrosion process and corrosion control. • understand the various methods of testing corrosion. • get idea on coating to prevent corrosion. • get familiar with corrosion type with respect to environment. • know about the concepts and applications of coating.						

Reference and Text Books: -

1. Bockris J.O.M, Conway B.E, Yeager E and White. (2013). *Electrochemical Materials Science in Comprehensive Treatise of Electrochemistry*, Volume 4. Plenum press.
2. Denny A. Jones. (2013). *Principles and Prevention of Corrosion*. Pearson.
3. Fontana M.G. (2017). *Corrosion Engineering*, McGraw Hill Education.
4. Hutchings I and Philip Shipwar. (2019). *Tribology: Friction and Wear of Engineering Materials*. Butter worth-Heinemann.
5. Sprowds D.O.(1986). *Corrosion Testing and Evaluation*. Corrosion Metals Hand book, Vol. 13.

II Semester							
DSE 2		**Elective**	**L**	**T**	**P**	**C**	**H/W**
Course code	**542512**	**SOLID STATE IONICS**	**-**	**T**	**-**	**3**	**3**
Objectives	The objective of the course is to make the students • knowthe basic aspects of solid state physics. • get knowledge on solid state ionics, hydrogen storage and nano-ionic materials. • learnaout micro batteries, fuel cells, super capacitors and their applications. • get familiar with various characterization techniques for new cathode materials. • learnabout various application of ionic materials.						
Unit I	**BASIC ASPECTS OF SOLID STATE PHYSICS -**Types of bonding in solids-Fundamentals of Crystallography-Simple Crystal structures-BCC, FCC, HCP - X -ray diffraction-band structures of metals, semiconductors and insulators-Ionicand electronic conductivities.						
Unit II	**SOLID STATE IONICS -** Concept of solid state ionics- Importance of super-ionic materials and structures -Classification of Superionic solids- crystalline anionic and cationic conductors, mixed ionic and electronic conductors-structural factors responsible for high ionic conductivity - Experimental probes pertaining to solid state ionics-Theoretical models of fast ion transport- Applications of fast ionic solids-Nano-ionic materials.						
Unit III	**MICRO BATTERIES AND APPLICATION –** Concept of a thin film solid state battery- electrolyte thin films - flash evaporation technique - electromotive force - reversible cells-free energy changes-capacity of a cell-power and energy density of a cell-polymer electrolytes-application of polymer electrolytes in micro batteries, Fuel cells-solid state battery-super capacitors.						
Unit IV	**CHARACTERIZATION OF NEW CATHODE MATERIALS** - Phase identification- Thermal analysis-DTA-DSC-TG- Energy dispersive X-ray fluorescence spectroscopy (EDX)-X-ray - X-ray photoelectron spectroscopy (XPS) - Structural characterization – XRD studies -Extended X-ray absorption fine structure - FTIR-Transport measurements.						
Unit V	**APPLICATIONS OF IONIC MATERIALS -**Primary lithium batteries-thermodynamics and mass transport in solid state batteries, battery performance and electrode kinetics-Secondary lithium batteries-Li-ion electrode materials-preparation and fabrication- -characterization of Li-ion cells- Comparison of Li- iodine and NiCd cells in CMOS-RAM applications. Applications of Lithium batteries.						
Outcome	After completion of the course, the students shall be able to • get familiar with basics of solid state physics. • know about the solid state ionics, hydrogen storage. • get familiar with micro-batteries, fuel cells, super capacitors. • get know about how cathode materials are characterized through various methods. • get familiar with applications of ionic materials						

Reference and Text Books: -

1. Chandra S. (1981). *Superionic Solids-Principles and applications*. North Holland Amsterdam.
2. Clive D. S. Tuck, (1991). *Modern Battery Technology*, Elis Horwood Publishers.
3. Crompton T.R. (2000). *Battery Reference Book*, Newnes.
4. Geoffrey, Ozin A. &Andre C Arsenault, (2008). *Nanochemistry: A Chemical Approach to Nanomaterials*, Royal Society of Chemistry.
5. Keer H.V. (2019). *Principles of the Solid State*. New Age International Private Limited.

III Semester							
DSE 3		**Elective**	**L**	**T**	**P**	**C**	**H/W**
Course code	**542514**	**BIOELECTRONICS**	**-**	**T**	**-**	**3**	**3**
Objectives	The objective of the course is to make the students • get knowledge on bioelectronic devices and their applications in various fields. • get knowledge on metal oxide-based semiconductor and its application in bioelectronic devices. • learn about molecular electronics • understand the microelectrodes and its behaviour. • get basic knowledge on bioelectricity and circuit models						
Unit I	**METAL OXIDE SEMICONDUCTOR (MOS) STRUCTURE -***pn* Junction, *pn*Junction Equilibrium, Effect of the Bias Voltage, Current – Voltage Characteristics of*pn* junction - MOS Structure - Accumulation Operating Mode- Depletion Operating Mode- Inversion Operating Mode, C-V Plots of an MOS Structure.						
UnitII	**METAL OXIDE SEMICONDUCTOR BASED BIOELECTRONIC DEVICES -** Biosensor overview- Transducers – characteristics - Ion sensitive field effect transistor – enzyme field effect transistor- Cell based biosensors and sensor of cell metabolism – light addressable potentiometric sensors (LAPS).						
Unit III	**MOLECULAR ELECTRONICS –** Molecular wires and switches; molecular arrays as memory stores biomolecular computer, Properties of DNA and its potential applications in molecular electronics.						
Unit IV	**MICROELECTRODES FOR BIOLOGICAL MONITORING** - Electrochemical cells, oxidation reduction reactions – Polarization, polarizable and non-polarizable electrodes, electrode behavior of circuit methods. Body surface recording electrode array, Microelectrodes for electric stimulation of tissues.						
Unit V	**BIOELECTRICITY AND BIOELECTRIC PHENOMENON -**Biology of the Neuron, Biophysical Description of the Action Potential, The Neuron as the threshold device – Synapses, Networks, Neuro-bioengineeringNeuroelectronic Junctions, Silicon Neurons Neurons - Equivalent Circuit Model for Cell Membrane - Hodgkin & Huxley and Equivalent Circuits.						
Outcome	After completion of the course, the students shall be able to • get familiar with metal oxide semiconductor and its structure. • learn about the role of metal oxide semiconductors in bioelectronic devices. • get basic knowledge in semiconductor interfacing with biomolecules towards bioelectronics devices. • get familiar with biological monitoring using microelectrodes. • learn about neuronal network and mechanism of signal transmission in biological system.						

Reference and Text Books: -

1. Andreas Offenhausserand Ross Rinaldi. (2009). *Nanobioelectronics-for electronics, biology and medicine*.
2. Chad A. Mirkin and Christ of Niemeyer M. (2007). *Nanobiotechnology II More Concepts and Applications.*
3. Christ of Niemeyer M and Chad A. Mirkin. (2004). *Nanobiotechnology Concepts, Applications and Perspectives.*
4. David S. Good sell. (2004). *Bionanotechnology*.
5. Jason J Davis. (2009). *Engineering the bioelectronic interface*.
6. Lagothetidis and Stergios. (2012). *Nanomedicine andNanobiotechnology*.
7. Oded Shoseyov, Ilan Levy. (2008). *Nanobotechnology.*

III Semester							
DSE 3		**Elective**	**L**	**T**	**P**	**C**	**H/W**
Course code	**542513**	**BIOSENSORS**	**-**	**T**	**-**	**3**	**3**

Objectives	The objective of the course is to make the students • gain knowledge in basic requirements to fabricate a sensor for a given application. • understand different methods for attaching recognition molecule on the sensor surface. • identify the interaction between the surface attached molecule and target molecule • understand the working principles of electronic and optical sensor devices. • know about the role of affinity sensors in disease diagnosis.
Unit I	**BASICS OF BIOSENSORS-**Biosensor – definition-Historical perspective; Sensor characteristics - calibration, dynamic Range, signal to noise, sensitivity, selectivity, interference- examples - applications –Problems.
Unit II	**TYPES OF TRANSDUCERS-** Transducer – definition- types – optical, electrochemical, Electrochemical transducers (Amperometric, potentiometric, conductimetric); - thermal, Mass – piezoelectric – acoustic wave with examples.
Unit III	**BIORECOGNITION SYSTEMS**– Enzymes; Microorganism based biosensor, immobilization of microorganism - botanical biosensors-Biosensors using cultured cells-intact tissues-receptor elements.
Unit IV	**DNA ELECTRONIC APPLICATIONS-** Molecular wires and switches Biomolecular computer, molecular arrays as memory stores, DNA for molecular devices - molecules between nanofabricated electrodes.
Unit V	**GLUCOSE SENSORS-** Definition- Historical developments – generations of glucose sensing -types of glucose monitoring – invasive and non-invasive – sensor market-Indian status.
Outcome	After completion of the course, the students shall • become knowledgeable in the field of biosensors. • understand the role of transducer in sensor and its types. • Know biorecognition system to detect particular type of analytes • Become familiar with the DNA sensors. • Gain knowledge on the history and recent development of glucose sensors.

Reference and Text Books: -

1. Cooper J. Cass T. (2004).*Biosensors*. 3- Biotechnology Advances.
2. Cooper J.M. Cooper J. Cass A.E.G. (2004). *Biosensors*. Oxford University Press.
3. Malhotra B.D. Turner A.P.F. (2003). *Advances in Biosensors*. Elsevier JAI.
4. Mulchandani A. Rogers K.R. (1998). *Enzyme and Microbial Biosensors Techniques and Protocols*. Humana Press, Totowa, New Jersey.
5. Zhang X. ZuH. Wang J. (2018). *Electrochemical Sensors, Biosensors and their Biomedical applications*. Elsevier Science and Technology Books.

III Semester							
DSE 3		**Elective**	**L**	**T**	**P**	**C**	**H/W**
Course code	**542515**	**CHEMICAL SENSORS**	**-**	**T**	**-**	**3**	**3**

Objectives	The objective of the course is to make the students • understand principles and application of modern chemical sensor technology. • getthe ability to operate with existing sensor systems and transducers, as well as to design new sensors, based on application of "smart materials". • learn about biochemical sensors. • get familiar with sensor engineering. • know more about application of sensor in various fields.
Unit I	**GENERAL PRINCIPLES, DEFINITIONS AND CONCEPTS -**Introduction to principles of chemical sensing; Signal transduction; Physico-chemical and biological transducers; Sensor types and technologies. Terminology and working vocabulary; Main technical definitions: calibration, selectivity, sensitivity, reproducibility, detection limits, response time; Problems and trade-offs.
Unit II	**PHYSICO-CHEMICAL SENSORS ANDTRANSDUCERS -** Thermal sensors; Electrochemical sensors (amperometric, potentiometric, conductimetric); Semiconductor transducers (ISFET, ENFET); Optical transducers (absorption, fluorescence, bio/chemiluminescence, SPR); Piezoelectric and acoustic wave transducers; Limitations & problems to be addressed.
Unit III	**BIOCHEMICAL SENSORS** – a. Enzymes; Oligonucleotides and Nucleic Acids; Lipids (Langmuir-Blodgett bilayers, Phospholipids, Liposomes); Membrane receptors and transporters; Immunoreceptors; Limitations & problems. b. Catalytic biosensors: mono-enzyme electrodes; bi-enzyme electrodes: enzyme sequence electrodes and enzyme competition electrodes. c. Affinity-based biosensors; Inhibition-based biosensors; Cell-based biosensors; Biochips and biosensor arrays; Problems and limitations.
Unit IV	**SENSOR ENGINEERING -** Methods for sensors fabrication: self-assembled monolayers, screen printing, photolithography, microcontact printing, MEMS. Engineering concepts for mass production.
Unit V	**APPLICATION -** Environmental monitoring; Technological process control; Food quality control; Clinical chemistry; Test-strips for glucose monitoring; Implantable sensors for long-term monitoring; Forensic science benefits; Problems & limitations.
Outcome	After completion of the course, the students shall be able to • get basic knowledge on principles, definition and concepts of sensors. • learn about transducers and its role in sensors. • get familiar with biochemical sensors. • get familiar engineering of sensors. • know about the application of sensors in various fields.

Reference and Text Books: -

1. Ajit S adanaand Neeti Sadana, (2011). *Handbook of Biosensors and Biosensor Kinetics*, Elsevier B.V. Amsterdam, The Netherlands. (ISBN: 978-0-444-53262-6)
2. Frieder S chelfer And Florian Schubert, (1992). *"Biosensors" Techniques And Instrumentation In Analytical Chemistry -Volume 11*, Elsevier Science Publishers B.V. Amsterdam, The Netherlands, (ISBN 0-444-98783-5).
3. Janata J. (2009). *Principles of Chemical Sensors*. Springer.
4. Jeong-Yeol Yoon, (2016). *Introduction to Biosensors: From Electric Circuits to Immunosensors*, Springer Int. Publishers (ISBN: 978-3-319-27411-9).
5. Jonathan M. Cooper and Cass A.E.G. (2004). *Biosensors: a practical approach*, Oxford University press. (ISBN 0-19- 9-63846- 2).
6. Vinod Kumar Khanna, (2011). *Nanosensors: Physical, Chemical, and Biological*, CRC Press. (ISBN: 978-1-439-82712-3).

III Semester							
DSE 3		**Elective**	**L**	**T**	**P**	**C**	**H/W**
Course code	**542516**	**THIN FILM SCIENCE AND TECHNOLOGY**	**-**	**T**	**-**	**3**	**3**

Objectives	The objective of the course is to make the students • know about mechanical pumps, production of high vacuum and thin film coating unit • getfamiliar with various methods for preparation of thin films. • understand the characterization methods used for thickness measurement. • gain knowledge on the nucleation theories and thin film structures. • gain knowledge on the various properties of thin films.
Unit I	**GROWTH AND STRUCTURE OF FILMS -**Introduction to thin films and applications - General features - Nucleation theories - Post-nucleation growth – Thin film structures- Structural defects.
Unit II	**THICKNESS MEASUREMENT AND MONITORING -** Multiple beam interference - quartz crystal - ellipsometric - stylus techniques. Characterization: X-ray diffraction - electron microscopy - high and low energy electron diffraction.
Unit III	**PREPARATION METHODS** – Physical methods: thermal evaporation - vapour sources - Wire, crucible and electron beam gun - sputtering mechanism and methods - Pulsed laser deposition (PLD), photochemical deposition (PCD) - Chemical methods: chemical vapour deposition and chemical solution deposition techniques - spray pyrolysis - laser ablation.
Unit IV	**PROPERTIES OF THIN FILMS** - Optical - reflection and anti-reflection coatings - interference filters - thin film solar cells – electro photography. Electrical and dielectric behaviour of thin films - components - thin film diode and transistor - strain gauges and gas sensors. Anisotropy in magnetic films - domains in films - computer memories - superconducting thin films - SQUID - mechanical properties: testing methods - adhesion - surface and tribological coatings.
Unit V	**HIGH VACUUM PRODUCTION-** Mechanical pumps - Diffusion pump - measurement of vacuum - gauges - production of ultra-high vacuum - thin film vacuum coating unit.
Outcome	After completion of the course, the students shall be able to • learn about growth and structure of thin films. • get familiar with measurement of thickness of thin films. • get familiar with various preparation methods of thin films. • get knowledge on properties of thin films. • get knowledge on production of high vacuum.

Reference and Text Books: -

1. Berry R.W, HallP.MandHarrisM.T. (1968). *Thin Film Technology*. Von Nostrand.
2. ChopraK.L. (1979). *Thin Film Phenomena*. Krieger Pub Co.
3. ChopraK.L. andKaurI. (2011). *Thin Film Device Applications*. Springer-Verlag New York Inc.
4. George Hass. (1963) *Physics of Thin Films: Volumes 1 -12*. Academic Press.
5. Goswami A.(2017). *Thin films Fundamentals*, New Age International (P) Ltd.
6. MaisselL.I. andGlangR. (Eds.). (1970). *Handbook of Thin film Technology*. McGraw- Hill.
7. OhringM. (2001). *Materials Science of Thin Films*. Academic Press.
8. Smith D. L. (1995). *Thin-Film Deposition: Principles and Practice*. McGraw-Hill.

III Semester							
DSE 3		**Elective**	**L**	**T**	**P**	**C**	**H/W**
Course code	**542517**	**SUPERCONDUCTING MATERIALS AND APPLICATIONS**	**-**	**T**	**-**	**3**	**3**

Objectives	The objective of the course is to make the students • understandthe basic experimental aspects of the superconductivity. • know about superconducting materials and its alloys. • understand the experimental studies of superconducting materials. • know the theoretical aspects of superconductivity. • learnabout various application in superconductivity.
Unit I	**BASIC EXPERIMENTAL ASPECTS -**Zero electrical resistance – Meissner effect – a.c. diamagnetic susceptibility – heat capacity – optical absorption by superconductor – entropy change –thermal conductivity – destruction of superconductivity by external magnetic fields – type I and type II materials – superconducting behaviour under high pressures –flux quantisation – normal and Josephson tunneling.
Unit II	**SUPERCONDUCTING MATERIALS -** Elemental superconductors –superconducting compounds and its alloys – A-I5 compounds – chevral phase compounds.
Unit III	**HIGH TEMPERATURE SUPERCONDUCTORS** – La-Ba-Cu-O, Y-Ba-cu-O, Bi-Sr-Ca-Cu-O and new systems and their crystal structures – Experimental studies on the new materials – organic superconductors –fullerenes.
Unit IV	**THEORETICAL ASPECTS -** Isotope effect – BCS theory – Role of electrons and phonons – applications of electron band structure results to calculate electron-phonon coupling constant McMillan's formula – GLAG theory– recent theories on high Tc materials, Coherence length, expression for critical temperature Tc, critical field Hc, critical current Jc – heavy fermion superconductivity.
Unit V	**APPLICATIONS -** Superconducting magnets – power generators, motors, transformers, power storage, power transmission – Josephson junction devices – IR sensors – SQUIDS –SLUGS – magnetically leviated trains – computer storage elements.
Outcome	After completion of the course, the students shall be able to • get familiar with basic concepts of superconductivity. • get knowledge in superconducting materials. • get familiar with high temperature superconducting materials. • get familiar with theoretical aspects of superconductivity. • learn about application of the superconductivity.

Reference and Text Books: -

1. Blundell S. (2009). *Superconductivity: A Very Short Introduction*. Oxford University Press.
2. Kowk H.S. andShawD.T (Eds.). (1988). *Superconductivity and its Applications*. Elsevier Science Publishing.
3. Narlikar A.V. (1990). *Studies on High temperature superconductors- Advances in research and applications*. Nova Scientific, New Delhi.
4. Narlikar A.V. andEkbote. (1983). *Introduction to Superconductivity*. South Asia publishers.
5. Schrieffer J.R. (2009). *Theory of Superconductivity*, Levant Books.
6. Tilley D.R. and Tilley. (1986). *Superfluidity and Superconductivity*. Adam Hilger.
7. Tinkham M. (2008). *Introduction to Superconductivity*. CBS Publishers & Distributors, New Delhi.

NON – MAJOR ELECTIVE

II Semester							
NME -II		**Non- Major Elective**	**L**	**T**	**P**	**C**	**H/W**
Course code	**542701**	**ELECTRONICS FOR DAILY LIFE**	**-**	**T**	**-**	**2**	**3**

Objective	The objective of the course is to make the students • get basics of electrical and electronics home appliances • learn about electrical accessories • get familiar with smart electronics • understand the concept of energy and energy devices • get knowledge on energy conservation
Unit I	**ELECTRICAL SAFETY -**General principles of electrical safety – Electricity and Human body - Electric shock and burn - Respiratory protection - Risk assessment and management - Safety against over voltage, extra-low and residual voltages - Hazardous areas, Electrical insulation - Electrical fires, Arc flash - Safety issues with emerging energy sources.
Unit II	**ELECTRICAL ACCESSORIES AND EARTHING -** Switches – holders – sockets – ceiling rose – plugs – main switch – fuse – circuit breaker – Earthing/grounding – importance – components of earthing system – types of earthing – pipe , plate and rod earthing – SI specifications of earthing.
Unit III	**SMART ELECTRONICS**– Historical Background of processor and Memory storage- Smart Phone, TAB, Laptop, Kindle – LCD and LED TV – smart watch- Medical diagnosis based on smart phone- Human–Computer Interaction.
Unit IV	**ENEGRY DEVICES**- Energy density vs Power density – Primary, Secondary Batteries- Wet Cell, Dry Cell- Alkaline-Lithium ion –Flow battery- Super capacitor- Fuel Cell.
Unit V	**ENERGY CONSERVATION-** Renewable Energy Source- Photovoltaic Cell – Energy Efficient lamps (CLF, LED)- Green Computing-Home appliance- Energy efficiency in Vehicles – Solar car.
Outcome	After completion of the course, the students shall be able to • get familiar with the principle of electrical safety. • get familiar in handling electrical appliances and electronics gadgets • learn about smart electronics. • get familiar with energy devices. • get familiar with how to conserve electricity by opting renewable energy sources and energy efficient home appliances.

Reference and Text Books: -

Albert Malvino, David J Bates.(2007).*Electronic Principles*, 7th Edition, McGraw Hill.
David A. Bell.(2007).*Electronic Devices and circuits*, 4th Edition, Prentice Hall.
Kishore, K. Lal.(2008).*Electronic Devices and circuits*, BS Publications; Third edition.
MehtaV.K. (2001).*Principles of Electronics*, 6th Revised Edition, S. Chandand Company.
PadiyarK.R. *Understanding the structure of electricity Supply*, B.S. Publications.
SzeS.M. (2008).*Semiconductor Devices: Physics and Technology*, Wiley India Pvt Ltd.

II Semester							
NME -II		**Non – Major Elective**	**L**	**T**	**P**	**C**	**H/W**
Course code		**FOOD CHEMISTRY**	**-**	**T**	**-**	**2**	**3**

Objectives	The objective of the course is to make the students • get knowledge on the macro and micro constituents of the food. • know the structure and chemical characteristics of constituents of food. • get knowledge of food chemistry and food systems. • Get familiar with the relationship between water and food. • Learn about rationale for certain food processes and preservation
Unit I	**INTRODUCTION TO FOOD AND ITS PROPERTIES -**Proteins-Enzymes-Chemistry and structure, kinetics, Maillard reaction. Food carbohydrates: Structural, nutritional and functional aspects. Emulsifiers-role of emulsifiers selection of emulsifier based on hydrophilic and Lipophilic balance (HLB) and its application. Thickeners-definition, chemical structure, gel formation, list of permitted thickeners and food application. Chemical and biochemical changes: changes occur in foods during different processing.
Unit II	**PROCESSING AND PRESERVATION -** Scope and benefits of industrial food preservation. Preservation of foods by chemicals, antibodies, antioxidants, salt and sugar. Principles of food freezing: freezing point of foods Psychrometric chart, Freeze concentration, freeze drying, IQF. Nanotechnology: Principles and application in foods, Hurdle technology: Types of preservation techniques and their principles, concept of hurdle technology and its application.
Unit III	**FLAVOURS AND COLOURING AGENTS** – Chemistry of food flavor, definitions, Flavourmatics /flavouring compounds, flavor retention-off flavours and food taints. Colour -Natural and synthetic food colours, their chemical structure, stability, permitted list of colours, usage levels and food application.
Unit IV	**WATER RELATIONS IN FOOD -** Moisture in food: Structure, properties, Types of water in food and their specific function water activity and stability.
Unit V	**FOOD ADDITIVES-** Definitions, uses and functions of: Acids, Bases, Buffer system, chelating/sequestering agents, Antioxidants, Anti-caking agents, Firming agents. Flour bleating agents and Bread improvers. Anti-microbial agents/ class I & II.
Outcome	After completion of the course, the students shall be able to • get basics about food quality and chemical constituents. • know about the techniques involved in food processing and preservation. • get familiar with importance of flavors and coloring agents in food. • learn about importance of food in food property and functions. • learn about functions of food additives.

Reference and Text Books: -

1. Belitz, H-D., Grosch, W. &Schieberle, P. (2004) *Food Chemistry 3rd Ed.* (translation of fifth German edition), Springer
2. Damodaran, S., Parkin, K. L., Fennema, O.R.(2008) *Fennema's Food Chemistry 4th Ed.* CRC Press
3. DeMan, J.M. (2018). *Principles of Food Chemistry 4rd Ed*. Aspen Publishers.
4. Harish Kumar Chopra and Parmjit Singh Panesar, (2010). *Food Chemistry*, Narosa Publication.
5. Jaswinder Kaur and Barry H. Grump.(2010). *Fundamentals of Food Chemistry*, Abhizeet Publications.
6. Peter C. K. Cheng, (2015). *Handbook of Food Chemistry*, Vol 1, Springer Reference.

III Semester							
NME -III		**Non – Major Elective**	**L**	**T**	**P**	**C**	**H/W**
Course code	**542702**	**NANOMATERIAL BIOSENSORS**	**-**	**T**	**-**	**2**	**3**

Objectives	The objective of the course is to make the students • understand basic characteristics of biosensors, nanoparticles and hybrids. • learn about Synthetic methods of nanoparticles in presence of biological molecules. • understand important biorecognition elements in biosensing. • know electrical and optical techniques in biosensing. • get familiar Biosensor applications in medical and food industries.
Unit I	**BASICS OF NANOBIOSENSORS -**Basic concepts, Classification, Components; Features of Biosensors – Sensitivity, Selectivity, Reproducibility, Portability, Stability, Detection Limit, Response time - types of nanobiosensors; Nanoparticle biomolecule - hybrids, Nanoparticle for biosensing.
Unit II	**BIOSYNTHESIS OF NANOPARTICLES -** Metal: Silver, gold, Platinum. Metal oxide: cerium, titanium, iron and zinc oxide nanoparticles - synthesis using bacteria, fungi, plant extracts, Biological applications of inorganic nanoparticles.
Unit III	**MOLECULAR RECOGNITION ELEMENTS IN NANOSENSING** – Transducers-bio-recognizing elements – Enzymes, Antibodies; Nucleic acids; Methods of Immobilization - Co-valent and non-covalent, self-assembly.
Unit IV	**ELECTRICAL AND OPTICAL BIOSENSORS -** Principles – Conductometric, amperometric and Impedimetric biosensors; Glucose biosensors - Optical Biosensors: Principles – Absorbance, Chemi-luminescence - Fluorescence, Phosphorescence; Colorimetric sensors.
Unit V	**NANOTECHNOLOGY AND ITS APPLICATION IN HEALTH AND FOOD INDUSTRY -** Nanotechnology and food packaging, natural biopolymers, advantages of nanomaterials in food packaging applications, nanosensors, outstanding issues, risks and regulations, public perception. Nanotechnology in Agriculture, Precision farming, Smart delivery system, Insecticides using nanotechnology, Potential of nano fertilizers.
Outcome	After completion of the course, the students shall be able to • get basic knowledge on nanomaterial biosensors. • get familiar with green synthesis of metal/oxide nanoparticles using bacteria, fungi and plant extracts. • get familiar with role of molecular recognition elements in nanosensig. • learn about electrical and optical properties for molecular sensing • get know about the applications of nanosensors in medical and food industries

Reference and Text Books: -

1. Charles P.Poole, Jr., FrankJ.Owens.(2006).*Introduction to Nanotechnology*, Wiley India.
2. Eltekhari, John Wiley Weinhim (2008).*Nanostructured Materials in Electrochemistry*.
3. Hari Singh Nalwa. *Nanostructured Materials and Nanotechnology*, Academic Press London USA, Concise Edition.
4. Kourosh Kalantar, Zadeh Benjamin Fry.(2008). *Nanotechnology Enabled Sensors*, Springer, Newyork.
5. Seminario, Jorge.(2014).*Design and applications of Nanomaterials for Sensor*, Springer Publications.
6. Vijay K.Vardan, L. Chen, Jining, John Wiley.(2008).*Nanomedicine Design and applications of Magnetic Nanoparticles*, Nanosensors andNanosystems, New Jersey.
7. William A. Goddard. *Handbook of Nanoscience and Technology*, CRC, Boca Raton, 2nd edition.

III Semester							
NME III		**Non Major Elective**	**L**	**T**	**P**	**C**	**H/W**
Course code	**NME**	**GREEN CHEMISTRY**	**-**	**T**	**-**	**2**	**3**
Objectives	The objective of the course is to make the students • learnthe basic concept and principles of green chemistry for environmental management. • know about green reagents and its importance to the environment • getfamiliar with green solvents and its impacts in green chemistry • get familiarize the synthesis of materials using green methods • get knowledge on applications of green synthesis technology						
UnitI	**PRINCIPLES OF GREEN CHEMISTRY -**History of green chemistry and sustainability- Prevention of waste/by-products – maximum incorporation of reactants in final product-Atom economy – Prevention/minimization of hazardous products – Designing safer chemicals – optimizing reaction conditions.						
UnitII	**GREEN REAGENTS AND CATALYSTS -** Choice of starting materials – reagents (Dimethyl carbonate, polymer supported reagents) – catalysts (microencapsulated Lewis acids, zeolites, basic catalysts polymer supported catalysts, introduction to biocatalysts).						
Unit III	**GREEN SOLVENTS** – Aqueous phase reactions (Claisen rearrangement, Aldol condensation, wurtz reaction, reduction of carbon carbon double bond, oxidation of amines into nitro compounds – Electrochemical synthesis (synthesis ofadiponitrile) **-** Ionic liquids – reactions in acidic ionic liquids- reactions in neutral ionic liquids (hydrogenations, diels-Alder reactions, Heck reactions, O-alkylation and N-alkylation, methylene insertion reactions.						
Unit IV	**GREEN SYNTHESES -** Microwave induced green synthesis (Hoffmann Elimination and Oxidation of alcohols) – Ultra sound assisted green synthesis (Esterification, Saponification andCannizaro reaction) – Solid state green synthesis (Dehydration of alcohols to alkenes, Grignard reaction)- Solid supported organic synthesis (Synthesis of furans and pyrrole).						
Unit V	**APPLICATIONS OF GREEN SYNTHESIS -** Introduction – synthesis of styrene, adipic acid, catechol, 3-Dehydroshikimic acid, methyl methacrylate, urethane. Environmentally benign synthesis of aromatic amines – free radical bromination – synthesis of ibuprofen and paracetamol.						
Outcome	After completion of the course, the students shall be able to • get familiar with basic concepts of green chemistry and apply them in various fields. • learn about catalytic reaction with green reagents and its importance. • identify available green solvents and apply them to various synthesis process • get familiar abou preparation of various drugs using green synthesis methods • apply skills and technology towards green chemistry and apply in industry.						

Reference and Text Books: -

1. Ahluwalia V. K. (2012). *Green Chemistry*, Narsoa publishers.
2. AhluwaliaV.K. andKidwaiM. (2004). *New trends in Green Chemistry*, Anamaya Publishers.
3. Bela Torokand Timothy Dransfield , (2017). *Green Chemistry, An Inclusive Approach*, 1stEdition, Elsevier.

CURRICULUM VITAE

Name : **Dr. C. SEKAR**
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Alagappa University
Karaikudi – 630003
Tamil Nadu,INDIA
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Educational qualification:

Ph.D. - Crystal Growth Centre, Anna University, Chennai.
M.Sc. - Pachaiyappa's College, University of Madras
B.Sc. - Sacred Heart College, University of Madras

Professional experience:

- Research – 14 Years
- Teaching – 21 Years

Area of Research:

- Materials Science: Metal oxide semiconductors, Carbon nanostructures, Biomaterials, Low dimensional cuprates
- Sensors: Chemical Sensors, Biosensors for Medical, Food, Agricultural and Environmental Applications

Honours and Awards:

- International Centre for Theoretical Physics (ICTP)-Italy (Travel grant-2015)
- Alagappa Excellence Award for Research(2016-17)
- SERB-DST, Govt. of India (International Travel Support – May2017)
- Best Poster Award –International Conference on Environmental Medicine, 10-11thDecember 2017, Kaohsiung Medical University, Taiwan.
- Visiting Researcher, University of Messina, Italy (May-July2017)
- International Centre for Theoretical Physics (ICTP)-Italy (Special grant-2019)

Recent publications:

- HAP-TiO_2 nanocomposites based electrochemical sensor for selective and simultaneous detection of para-aminohippuric acid and uric acid, *Microchemical Journal* (2022).
- Precise and quick detection of ascorbic acid and eugenol in fruits, pharmaceuticals and medicinal herbs usingHAP-TiO_2 nanocomposites -based electrode, *Food Chemistry* (2022).
- Pseudo spin-ladder CaCu2O3 nanostructures as potential electrode material for asymmetric supercapacitors, *Journal of Energy Storage* (2022).
- Highly efficient non-enzymatic electrochemical determination of histamine based on tungsten trioxide nanoparticles for evaluation of food quality, *Journal of Applied Electrochemistry* (2021).
- Shock waves exposed α-Fe2O3 nanoparticles for electrochemical sensing of riboflavin, uric acid and folic acid. *Microchemical Journal* (2021).

Total Publications: 149, **Total Citation:** 3166, **h- index:** 31, **i10- index:** 68

CURRICULUM VITAE

Name : **Dr. Giovanni Neri**
Designation : Professor
Address : Department of Engineering
University of Messina.
Italy.
Phone : 090 6765297
Email : gneri@unime.it

Educational qualification:

M. S Chemistry – University of Messina, Italy.

Area of Research:

- Catalysis, Gas sensors, Biosensors and Material Science

Honours and Awards

- Awarded a grant by Samsung SAIT Global Research Outreach program for the project
  Smart sensors for breath analysis.
- Visiting researcher/professor at the College of Chemical Engineering, The University of Michigan
  (1991 and 1996),
- Visiting researcher/professor at Alagappa University (2013 and 2019).
- Visiting researcher/professor at Indian Institute of Technology (IIT) Indore (2016).
- Associate Editor-in-Chief of Chemosensors, member of the Editorial Board of Sensors and Academic Editor for the "Gas Sensors" topical collection.

Recent publications:

- Role of Hetero junctions of Core–Shell Hetero structures in Gas Sensing, *ACS Applied Materials & Interfaces* (2022).
- Low temperature detection of nitric oxide by CuO nanoparticles synthesized by pulsed laser ablation, *Sensors and Actuators B: Chemical* (2022).
- A Capacitive Sensor, Exploiting a YSZ Functional Layer, for Ammonia Detection, *IEEE Transactions on Instrumentation and Measurement* (2022).
- A novel yttria-doped ZrO2 based conductometric sensor for hydrogen leak monitoring, *International Journal of Hydrogen Energy* (2022).
- Development of a novel electrochemical nitrite sensor based on Zn-Schiff base complexes, *Applied Organometallic Chemistry* (2022).

Total Publications: 476, **Total Citation:** 16184, **h- index:** 59, **i10- index:** 196

CURRICULUM VITAE

Name : Dr. N. Ramesh Babu

Designation : Associate Professor

Address : National Institute of Technology
Tiruchirappalli.

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Educational qualification:

- Ph.D. (Biomaterials-Metallurgical and Materials Engineering), Indian Institute of Technology Madras, 2007
- M.Tech. (Metallurgical Engineering), Indian Institute of Technology Madras, 2001
- A.M.I.I.M (Metallurgical Engineering), Indian Institute of Metals Kolkata, 1997

Area of Research:

Bio materials, Ceramic materials and Plasma electrolytic oxidation coatings.

Honours and Awards:

- Sudharsan Bhat Memorial Prize for the Best Ph.D. Thesis in Metallurgical and Materials Engineering, IIT Madras (2007).
- Best paper award for the paper presented at International Conference on Advances in Surface Treatment: Research and Applications (ASTRA-2003), Nov 3-6, 2003, Hyderabad.
- Best paper award for the paper presented at International Conference on Future Bioengineering (ICFB 2012) in Bangkok, Thailand, November 24-25, 2012.
- Best paper award in International Conference on Processing and Fabrication of Advanced Materials (PFAM 28) in the category Surface Coatings, 7-8th Dec 2020
- Best paper award in International Conference on Processing and Fabrication of Advanced Materials (PFAM 28) in the category Alloys and compounds, 7-8th Dec 2020)

Recent publications:

- Development of immobilised sunlight active W-Mo/Mo-V/VW co-doped TiO_2 photocatalyst by plasma electrolytic oxidation, *Journal of Alloys and Compounds* (2022).
- Facile preparation of immobilised visible light active W–TiO2/rGO composite photocatalyst by plasma electrolytic oxidation process, *Physica B: Condensed Matter* (2022).
- Laser Sintered Ti-6Al-4V Alloy Coated with Plasma Electrolytic Oxidation: Influence of Duty Cycle and Frequency on Morphological, Structural, and Corrosion Properties, *Journal of Materials Engineering and Performance* (2022).
- Effect of Sodium Aluminate in the Borax-Based Electrolyte on Corrosion Resistance of Plasma Electrolytic Oxidation Coatings Fabricated on Galvanized Steel, *Transactions of the Indian Institute of Metals* (2022).
- Superior properties and behaviour of coatings produced on nanostructured titanium by PEO coupled with the EPD process, *Surface Topography: Metrology and Properties* (2022).
- Effect of swift heavy silicon ion irradiation on TiO2 thin film prepared by micro arc oxidized technique, *Materials Today: Proceedings* (2022).

Total Publications: 90, **Total Citation:** 3344, **h- index:** 32, **i10- index:** 49

CURRICULUM VITAE

Name : **Dr. J. Mathiyarasu**
Designation : Senior Principal Scientist
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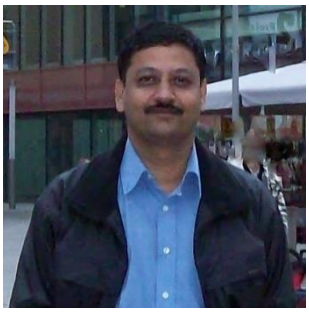

Educational Qualification:

M.Sc., B.Ed., Ph.D.

Area of Research:

Electrochemical biosensors, Nanomaterials, Polymers, Material Science and Micro fabrication

Honours and Awards:

- BOYSCAST Fellowship (2007-08) Department of Science and Technology, New Delhi
- Visit to department of Materials Science, The Angstrom Laboratory, Uppsala University, Uppsala, Sweden.
- Visit to Kurt-Schwabe Institute for Sensor and Measuring Technology, Meinsberg, Germany for discussions in connection with an Indo-German (CSIR-FzJ) programme of co-operation in science & technology entitled "Design and Development of Electrochemical Sensors for Human Health-care Applications"

Recent publications:

- Environmental-assisted shape-controlled synthesis and electro catalytic performance of CuS nanostructures for vanillin detection in commercial food products, *Applied Materials Today* (2022).
- Self-Healing Gel Electrolytes for Flexible Supercapacitors, *Polymers in Energy Conversion and Storage* (2022).
- Electrodeposited partially oxidized Bi &NiCo alloy based thin films for aqueous hybrid high energy microcapacitor, *Journal of alloys and compounds* (2021).
- A highly uniform CuO@ SiO2 porous sphere with improved electrochemical sensing performance for the accurate determination of vanillin in food samples, *Materials Today Chemistry* (2021).
- Detection of white spot syndrome virus in seafood samples using a magnetosome-based impedimetric biosensor, *Archives of Virology* (2021).

Total Publications: 91, **Total Citation:** 3271, **h- index:**32, **i10- index:**61

CURRICULUM VITAE

Name : **Dr. V. Viswabaskaran**
Designation : Ceramic Consultant
Address : VB ceramics,
Kottivakkam, Chennai, India.
Email :drvbcrc@gmail.com

Educational qualification:

M.Tech., Ph.D

Area of Research:

Ceramic Research

Professional Experience:

Teaching – 6 Years
Industry – 10 Years

Honours and Awards:

- Winner of Amikaward for his best performance in the Associate membership exam conducted by Indian Institute of Ceramics, Kolkata.
- Life member in Indian Ceramic Society and Indian Institute of Ceramics, Kolkata.
- GANPULE AWARD from The Indian Ceramic Society.
- PROF. SASADHAR RAY MEMORIAL award for industrial excellence from ICS
- INDIAN ACHIVERS AWARD FOR INDUSTRIAL DEVELOPMENT from Indian Achievers' Forum

Total Publications: 17

CURRICULUM VITAE

Name : **Dr. G. RAVI**
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Address : Department of Physics
Alagappa University
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Tamil Nadu, INDIA
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Educational qualification:

- Ph.D. - Anna University, Chennai
- M.Phil. - Anna University, Chennai
- M.Sc. - Bharathidasan University
- B.Sc. - Bharathidasan University

Professional experience:

- Research – 27 Years
- Teaching – 23 Years

Area of Research:

- Crystal growth of organic & inorganic materials
- Nano materials synthesis and Thin Films preparation for super capacitors, Photo catalytic and sensor applications
- Opto-electronics and E-O modulator–Devices

Honours and Awards:

- Visiting Professor, Shizuoka University, Japan, Aug-Nov. 2012
- Honorable Guest Professor, Shizuoka University, Japan, April2014
- Alagappa Excellence Award for Research (2015-2016), Alagappa University, 2016
- Honorable Guest Professor, Shizuoka University, Japan, April2016
- JSPS Invitation Fellowship, Japan, Nov.-Dec. 2016
- Appreciation Award, Alagappa University, Karaikudi, Feb. 2017
- Honorable Guest Professor, Shizuoka University, Japan, April 2018 17.
- Honorable Guest Professor, Shizuoka University, Japan, April2019

Recent publications:

- Synthesis of self-assembled micro/nano structured manganese carbonate for high performance, long lifespan asymmetric super capacitors and investigation of atomic-level, *Journal of Energy Storage* (2020).
- Synthesis of X 3(PO4)2 [X = Ni, Cu, Mn] Nanomaterials as an Efficient Electrode for Energy Storage Applications, *Journal of Nanoscience and Nanotechnology* (2020).
- Neutral and alkaline chemical environment dependent synthesis of Mn3O4 for oxygen evolution reaction (OER), *Materials Chemistry and Physics* (2020).
- Fabrication and electrochemical OER activity of Ag doped MoO3 nanorods, *Materials Science in Semiconductor Processing* (2020).
- Super capacitor and OER activity of transition metal (Mo, Co, Cu) sulphides, *Journal of Physics and Chemistry of Solids* (2020).

Total Publications: 295, **Total Citation**: 3434, **h- index**: 31, **i10- index**: 91

CURRICULUM VITAE

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Email :dharumanudhay@yahoo.com

Educational qualification:

- Ph.D. - University of Madras
- B.Ed. -University of Madras
- M.Sc. - University of Madras
- B.Sc. - University of Madras

Professional experience:

- Research – 15 Years
- Teaching – 9 Years

Area of Research:

Chemistry/Electrochemistry/ Diabetic, cancer biosensors development using, DNA, antibody (immunosensors) and neurological disorder sensors

Honours and Awards:

- Best poster award, National Conference on Futuristic Materials (NCFM-2017),AlagappaUniversity, Karaikudi
- Best poster award, International conference on recent advance in materials and chemical sciences (ICRAMCS-2015), Gandhigram Rural Institute
- Alagappa Excellence Award for Research2015-2016
- Best poster award, Indo-Japan workshop on Biomolecular Electronics & Organic Nanotechnology for Environment Preservation (IJWBME 2013), Delhi Technological University, Delhi,India
- Young Biomedical scientist Research Fellowship by Indian Council of Medical Research, for the year2012-2013
- Article Gold nano particle decorated graphene core first generation PAMAM dendrimer for label free electrochemical DNA hybridization sensing, Biosens. Bioelectr., 31 (2012) 406-412. Ranked 16th on the TOP 25 articles in the Journal of Biosensors and Bioelectronics, March2012
- Research Scientist, AIST, Japan, Oct.2006 – March2007
- Brain Korea Post-doctoral research fellowship, December 2004

Recent publications:

- Physicochemical and electrochemical analysis of rare earth metal doped BTO perovskite thin films, (2019).
- Carbon dots stabilized silver-lipid nano hybrids for sensitive label free DNA detection, (2019).
- Carbon dots stabilized silver–lipid nano hybrids for sensitive label free DNA detection, (2019).
- Single step sol-gel synthesized Mn_2O_3-TiO_2 decorated graphene for the rapid and selective ultra sensitive electrochemical sensing of dopamine, (2018).
- Self-powered polymer–metal oxide hybrid solar cell for non-enzymatic potentiometric sensing of bilirubin, (2018).

Total Publications: 58, **Total Citation:** 947, **h- index:** 18, **i10- index:** 24

CURRICULUM VITAE

Name : **Dr. J. Wilson**
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Address : Department of Bioelectronics& Biosensors
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Educational qualification:

- Ph.D. - Alagappa University Karaikudi.
- M.Sc. -St. Joseph's College Bharathidasan University Trichy
- B.Sc. - Alagappa University Karaikudi

Professional experience:

- Research – 14 Years
- Teaching –8Years

Area of Research:

- Conducting polymers, Metal oxides, Carbon based materials, Biosensors, lithium batteries

Honours and Awards:

- Nature India has highlighted our work, doi:10.1038/nindia.2016.27

Recent publications:

- Sensitive voltammetric sensor based on silver dendrites decorated polythiophene nanocomposite: Selective determination of L-Tryptophan (2020).
- Size controllable, pH triggered reduction of bovine serum albumin and its adsorption behavior with SnO_2/SnS_2 quantum dots for biosensing application (2020)
- Solvothermal synthesis of magnetically separable reduced graphene oxide/Fe3O4 hybrid nano composites with enhanced photo catalytic properties(2020)
- Mesoporous nickel oxide nanostructures: Influences of crystalline defects and morphological features on mediator free electrochemical monosaccharide sensor application (2019).
- Non – Enzymatic L – Tyrosine Detection Based on PEDOT/ZrO_2 /rGO Composite (2019).

Total Publications: 37, **Total Citation:** 919, **h- index:** 13, **i10- index:** 19

CURRICULUM VITAE

Name : **Mrs. S. Meenakshi**
Designation : Research Scholar
Address : Alagappa University
karaikudi, India.
Email : meenakshisolaimani@gmail.com

Area of Research:

Chemo Biosensors, Shock waves

Honours and Awards:

- ANVESHAN 2019-2020 – got Second position in the **zonal level**- Student research convention on the project entitled "Electrochemical Sensor for Quick Non-invasive Diagnosis of Renal and Hepatic Disorders" held at Bharathidasan University.
- ANVESHAN 2019-2020 – got First position in the **National level**- Student research convention on the project entitled "Electrochemical Sensor for Quick Non-invasive Diagnosis of Renal and Hepatic Disorders" held at RGPV,Bhopal.

Recent publications:

- Shock waves exposed α-Fe_2O_3 nanoparticles for electrochemical sensing of riboflavin, uric acid and folic acid
- Precise and quick detection of ascorbic acid and eugenol in fruits, pharmaceuticals and medicinal herbs using hydroxyapatite - titanium dioxide nanocomposite-based electrode

Total Publications: 2